Fast and Robust Phase Equilibrium Computations for CO_2 + H_2O + NaCl Mixtures Using the Electrolyte Cubic-Plus-Association Equation of State

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Abstract

We present phase stability and flash calculations for CO_2 + H_2O + NaCl mixtures using the electrolyte Cubic-Plus-Association (eCPA) equation of state parameterized by Coelho et al. (2025). In this ternary system, H_2O partitioning into the CO_2-rich phase concentrates NaCl in the aqueous phase, raising the equilibrium molality m_s^{aq} above the feed value and suppressing CO_2 solubility — a salting-out coupling that makes phase compositions feed-fraction-dependent and requires full stability+flash rather than a fixed VLE lookup. Starting from the eCPA thermodynamic framework, we develop: (i) a generalized salt-free CPA flash routine with Michelsen tangent-plane distance (TPD) stability tests using six initial guesses and accelerated successive-substitution iterations (SSI) following Jex et al. (2024), validated for the CO_2 + H_2O binary over 29,412 conditions with 100% convergence; (ii) an extension to the full ternary system via analytical elimination of m_s^{aq} within each SSI step; (iii) a precomputed three-dimensional solution table spanning $T = 0$–$360\,^\circ\mathrm{C}$, $P = 1$–$2000\,\mathrm{bar}$, and $m_s = 0$–$6\,\mathrm{mol\,kg^{-1}}$ (505,400 grid points) that provides warm-start initial guesses reducing flash cost by 3.2× for the ternary system and more than two orders of magnitude for the salt-free CPA binary (cold-start six-trial robust algorithm versus warm-started SSI); and (iv) a simplified flash exploiting $y_{\mathrm{H_2O}} \approx 0$, valid for $T \lesssim 80\,^\circ\mathrm{C}$ with errors below 2% in dissolved CO_2 molality. Accelerated SSI reduces the mean flash iteration count by 46% (25.4 to 13.7) versus standard SSI; analytical Newton polish after warm-started SSI further reduces the mean to 6.9 iterations (∼20% wall-time saving). Analytical Jacobian-based Newton–Raphson inner solvers reduce the per-iteration cost of CPA/eCPA to only ∼3× that of a plain cubic EoS, and are 6–13× faster than cold-start alternatives. The six-guess stability test detects all two-phase conditions without false negatives. Validation against >1100 experimental data points achieves 100% flash convergence with average absolute relative errors (AARE) of 9.6% for CO_2 solubility in pure water (6.5% under subsurface storage conditions) and 24.4% for H_2O content in the CO_2-rich phase (17.4% after excluding flagged outlier datasets); 6.6% for CO_2 molality in NaCl brine and 6.2% across all converged points ($T = 5$–$350\,^\circ\mathrm{C}$); and 0.76% for aqueous-phase density without any volume shift correction for water.

Keywords: CO_2 sequestration, eCPA equation of state, phase stability, flash calculation, accelerated successive substitution, reservoir simulation

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1. Introduction

Geological storage of CO_2 in saline aquifers and depleted hydrocarbon reservoirs is one of the most important near-term options for large-scale carbon capture and storage (Metz et al., 2005; Orr, 2004). Geothermal energy extraction presents an equally compelling application (Solovský and Firoozabadi, 2026): enhanced geothermal systems and hydrothermal reservoirs operate at temperatures often above 227 °C, where CO_2 can serve as a working fluid or co-migrate with brine, and where the phase behavior of the CO_2 + H_2O + NaCl system must be accurately described to model fluid circulation, energy extraction efficiency, and mineral scaling. In both applications, accurate modeling of CO_2+brine phase behavior is a prerequisite for reliable reservoir simulation: the equilibrium compositions of the CO_2-rich and aqueous phases govern injectivity, trapping efficiency, and caprock integrity, while accurate density predictions are needed for fluid flow and pressure calculations. The high-temperature range ($T > 227\,°\mathrm{C}$) is particularly relevant to geothermal applications, motivating the wide temperature coverage of the present model.

The equilibrium of CO_2 with brine is challenging to model. It spans a wide range of conditions—temperatures from near-ambient (∼10 °C) to supercritical and hydrothermal (>327 °C), pressures from hydrostatic (1 bar) to lithostatic (>1000 bar), and salinities from fresh water to near-saturation (>6 mol kg^{-1} NaCl)—and involves both association interactions (hydrogen bonding in water), CO_2–water cross-association, and electrostatic effects of dissolved ions.

Cubic-Plus-Association (CPA) equations of state (Kontogeorgis et al., 1996) successfully combine a cubic (Soave–Redlich–Kwong, SRK) repulsive/dispersion term with a Wertheim association term (Wertheim, 1984) and have been widely applied to systems with hydrogen-bonding fluids. Considering the *cross*-association between CO_2 and H_2O molecules further improved the predictive powers of CPA considerably (Li and Firoozabadi, 2009). The electrolyte CPA (eCPA) extension (Maribo-Mogensen et al., 2015) adds Debye–Hückel and Born electrostatic terms to describe ion–solvent and ion–ion interactions. Coelho et al. (2025) recently parameterized eCPA for the CO_2 + H_2O + NaCl ternary system with temperature-dependent binary interaction parameters, achieving a 74% reduction in CO_2 solubility error in brine compared to the prior model of Maribo-Mogensen et al. (2015), which neglected cross-association (AARE: 6.9% vs. 26.6% over 660 data points from 10 to 450 °C).

Despite the availability of well-parameterized thermodynamic models, practical use in reservoir simulation has been limited by the lack of open, efficient phase equilibrium routines for multicomponent mixtures. A key thermodynamic subtlety compounds this challenge. For a binary mixture such as CO_2 + H_2O, the Gibbs phase rule fixes the compositions of the coexisting phases at any given T and P: only the phase *amounts* depend on the overall CO_2 feed mole fraction, so a simple vapor–liquid equilibrium (VLE) table indexed on (T, P) fully characterizes the system. In the ternary CO_2 + H_2O + NaCl system this no longer holds. As CO_2 partitions into the CO_2-rich phase it carries along some water, so the equilibrium aqueous NaCl molality m_s^{aq} exceeds the feed molality m_{feed} by an amount that grows with the CO_2 phase fraction β—the well-known salting-out effect. Because m_s^{aq} depends on the overall compositions, $\mathbf{z}$, through the phase split, the equilibrium phase compositions themselves depend on $\mathbf{z}$: a VLE table at fixed feed salinity is thermodynamically inconsistent for the ternary system, and full stability+flash is required to resolve the self-consistent (β, m_s^{aq}, $\mathbf{x}$, $\mathbf{y}$) output for every $(T, P, \mathbf{z}, m_{\mathrm{feed}})$ query. Reservoir simulators therefore need flash calculations that are both *robust*—returning the correct phase split across the entire physically relevant parameter space—and *fast*—executing in milliseconds per grid cell to be compatible with time-stepping over millions of cells. Achieving this for an EoS as complex as eCPA (which requires iterative solution of both the cubic compressibility equation and the nonlinear association equations) is non-trivial.

This work makes the following contributions:

1. A complete implementation of the full eCPA equation of state of Coelho et al. (2025), including Debye–Hückel, Born, association, and permittivity terms, validated against the original parameterization over $T = 10$–$455\,°\mathrm{C}$.
2. Analytical Jacobian-based Newton–Raphson solvers for both the CPA and eCPA equations of state (EOS) that reduce the computational cost to only $\sim 3\times$ that of a plain cubic EoS with an analytical Z-root solution, rather than the one-to-two orders of magnitude typically assumed.
3. A generalized salt-free CPA flash routine with Michelsen tangent-plane distance (TPD) stability testing (Michelsen, 1982) using six initial guesses (Jex et al., 2024) and accelerated successive-substitution iteration (SSI) with dominant-eigenvalue step-size control (Jex et al., 2024), validated for the CO_2 + H_2O binary across 29,412 (T, P, z) conditions with 100% convergence and a $\sim 2\times$ iteration-count reduction over standard SSI.
4. An extension to the ternary CO_2 + H_2O + NaCl system, in which the equilibrium aqueous molality m_s^{aq} is determined self-consistently within the SSI loop, replacing an earlier robust but slow outer root-finding loop over molality.
5. A precomputed three-dimensional $(T,\ P,\ m_s)$ solution table with 505,400 grid points covering $T = 0$–$360\,°\mathrm{C}$, $P = 1$–$2000\,\mathrm{bar}$, and $m_s = 0$–$6\,\mathrm{mol\,kg^{-1}}$, computed with the full hierarchical stability+flash algorithm (100% convergence), and a fast warm-started flash that reduces iteration counts from ~ 12 to ~ 3 ($3.2\times$ wall-time speedup), suitable for integration into reservoir simulators.
6. Several algorithmic optimizations: multi-core parallelization for solution-table construction, phase-envelope tracing, and stability mapping; temperature and salinity continuation within the table build to maximize convergence rates; and a logarithmic pressure axis for the interpolation grid.
7. Systematic validation against experimental vapor–liquid equilibrium (VLE) and density data, demonstrating 0.76% AARE for aqueous-phase density without a volume shift correction for water—a strength of the CPA equation of state.
8. A minimal but complete IMPEC compositional reservoir simulator (both Mixed Finite Element and Two-Point-Flux-Approximation options, two-phase Darcy flow, Corey relative permeabilities) that integrates the fast CPA and eCPA flash routines and demonstrates 8,840–14,270 flash calls $\mathrm{s^{-1}}$ on a 2500-cell grid (80–87% of step cost), with a discussion of how the routine can be embedded in existing simulators such as MRST (Lie, 2019), OPM Flow (Rasmussen et al., 2021), and DuMu^x (Koch et al., 2021).

The remainder of this paper is organized as follows. Section 2 presents the CPA and eCPA thermodynamic framework. Section 3 describes the phase stability test and flash algorithms, including the hierarchical multi-guess strategy, the accelerated SSI, the precomputed solution table, and a simplified flash exploiting $y_{\mathrm{H_2O}} \approx 0$. Section 4 validates the implementation against experimental CO_2 + H_2O and CO_2 + H_2O + NaCl VLE and density data. Section 5 analyzes computational performance of the inner and outer Newton–Raphson solvers and the warm-start speedup. Section 6 demonstrates the flash routines within a minimal IMPEC reservoir simulator. Conclusions are given in Section 7. Some additional technical details and supporting figures are provided in Supplemental Information.

2. Thermodynamic Framework

Throughout this work species are indexed $H_2O \equiv 1$, $Na^+ \equiv 2$, $Cl^- \equiv 3$, $CO_2 \equiv 4$; for the salt-free CPA binary, indices 2 and 3 are absent and only species 1 (H_2O) and 4 (CO_2) are present. NaCl is treated as a single molecular species (with ionic parameters for Na^+ and Cl^- entering the electrostatic terms); extension to multiple-salt brines on an ionic basis is discussed in Section 7.

Mole fractions in the aqueous phase are denoted x_i and in the CO_2-rich phase y_i, with equilibrium ratios (K-values) $K_i = y_i/x_i$. Where it aids readability, explicit subscripts are used instead of indices (e.g. $x_{H_2O} \equiv x_1$, $y_{CO_2} \equiv y_4$); the two notations are used interchangeably. A complete list of symbols is given in the Nomenclature section of the Supporting Information.

2.1. CPA equation of state

For the salt-free CO_2 + H_2O binary the residual Helmholtz energy per mole is

$$A^r = A^{\mathrm{SRK}} + A^{\mathrm{assoc}}, \tag{1}$$

where A^{SRK} is the Soave–Redlich–Kwong (Soave, 1972) physical term and A^{assoc} is the Wertheim association term (Wertheim, 1984).

The SRK compressibility factor satisfies

$$Z = \frac{v}{v-b} - \frac{a(T)}{RT}\frac{1}{v+b}, \tag{2}$$

where v is the molar volume and x_i is the mole fraction of species i; $a(T) = \sum_i \sum_j x_i x_j (1-k_{ij})\sqrt{a_i a_j}$ and $b = \sum_i x_i b_i$ are the mixture attractive and co-volume parameters, respectively, with sums over molecular species $i, j \in \{H_2O,\ CO_2\}$ (and additionally Na^+, Cl^- for brines). The Soave temperature dependence is $a_i(T) = a_{0i}[1 + c_{1i}(1 - \sqrt{T/T_{c,i}})]^2$. The Peng–Robinson (PR) equation of state (Peng and Robinson, 1976) can serve equally well as the cubic backbone; it has been used with cross-association in the CPA framework for water-containing mixtures (Li and Firoozabadi, 2009; Moortgat, 2018; Nasrabadi et al., 2016; Moortgat, 2025; Moortgat et al., 2011a). We follow Coelho et al. (2025) in using SRK throughout the present work.

The association term accounts for hydrogen bonding in water via the 4C scheme (two proton donors and two proton acceptors), and includes cross-association between CO_2 and H_2O (Li and Firoozabadi, 2009). The sum runs over molecular species only ($i \in \{1, 4\}$); ions carry no association sites:

$$\frac{A^{\mathrm{assoc}}}{NRT} = \sum_i x_i \sum_{A\in i} \left(\ln \chi^{A_i} - \frac{\chi^{A_i}}{2} + \frac{1}{2} \right), \tag{3}$$

where χ^{A_i} is the fraction of molecules i not bonded at site A:

$$\chi^{A_i} = \frac{1}{1 + c\sum_j x_j \sum_{B\in j} \chi^{B_j}\, \delta^{A_i B_j}}, \tag{4}$$

with $c = N_A/v$ the molecular number density (N_A is Avogadro's number, v the molar volume) and $\delta^{A_i B_j}$ the association bonding volume between site A on species i and site B on species j. The CO_2–H_2O cross-association strength is proportional to water self-association (Coelho et al., 2025):

$$\delta^{A_w B_c} = S_{cw}(T)\, \delta^{A_w B_w}, \tag{5}$$

where $S_{cw}(T)$ and the binary interaction parameter $k_{cw}(T)$ both follow quadratic polynomials in $T/T_{c,c}$ ($T_{c,c} = 31.3\,°\mathrm{C}$, the CO_2 critical temperature), with coefficients from Coelho et al. (2025) (Table B.10).

The temperature dependence of both k_{cw} and S_{cw} is essential because the CO_2+H_2O system spans qualitatively different regimes across the validated temperature range. Below $T_{c,c} = 31.3\,°C$, CO_2 is a subcritical liquid and its fugacity is governed primarily by vapor-pressure equilibrium. Between $\sim$31 °C and $\sim$177 °C, CO_2 is supercritical and the aqueous solubility decreases steeply with temperature. Above $\sim$227 °C, the CO_2-rich phase acquires substantial water content (up to 20–25 mol% at $T = 227\,°C$ and $P = 200\,\text{bar}$), making accurate modeling of both phases equally important. A single set of temperature-independent binary interaction parameters cannot simultaneously describe all three regimes; the quadratic polynomial form of Coelho et al. (2025) provides sufficient flexibility while remaining analytically tractable.

2.2. eCPA equation of state

For the ternary system, two electrostatic terms are added (Maribo-Mogensen et al., 2015):

$$A^r = A^{\text{SRK}} + A^{\text{assoc}} + A^{\text{DH}} + A^{\text{Born}}, \tag{6}$$

where A^{DH} and A^{Born} are the Debye–Hückel and Born solvation contributions, respectively. The presence of NaCl (Na^+ + Cl^- ion pair) modifies the dielectric constant ε_r of the aqueous phase, which enters both electrostatic terms. The permittivity is computed self-consistently from the Onsager–Kirkwood equation, linking it to the Wertheim association fractions and the molecular dipole moments (Coelho et al., 2025).

Ion–molecule interactions in the SRK mixing rule are modeled via energy parameters ΔU_{is} of the Non-Random Two-Liquid (NRTL) type with a temperature dependence of the form

$$\Delta U_{is} = \Delta U_{is}^{\text{ref}} + \alpha_{is} R\left[\left(1 - \frac{T}{T_{is}^{\alpha}}\right)^2 - \left(1 - \frac{T_{\text{ref}}}{T_{is}^{\alpha}}\right)^2\right], \tag{7}$$

with $T_{\text{ref}} = 25\,°C$ and parameters from Coelho et al. (2025). All EoS parameters are listed in Tables B.9 and B.10.

2.3. Density and Péneloux volume correction

The molar volume in each phase is $v = ZRT/P$. Cubic equations of state systematically over- or underpredict liquid densities. The Péneloux volume translation (Péneloux et al., 1982) corrects this without affecting phase compositions (it is isofugacity-preserving):

$$v^{\text{corr}} = v^{\text{EoS}} + \sum_i x_i c_i, \tag{8}$$

with the corrected density $\rho = M_{\text{mix}}/v^{\text{corr}}$. Coelho et al. (2025) apply a shift $c_s = -53.5\,\text{cm}^3\,\text{mol}^{-1}$ to the NaCl species to correct brine density. No volume shift is applied to the water or CO_2 components; as shown in Section 4.4, the uncorrected eCPA model already achieves sub-1% accuracy for aqueous-phase densities.

3. Phase Stability and Flash Computations

3.1. Initial implementation: outer Brent loop over aqueous molality

As a first, robust implementation of the eCPA flash, we employ an outer Brent root-finding loop over the equilibrium NaCl molality m_s^{aq} [$\text{mol}\,\text{kg}^{-1}$] in the aqueous phase. The Brent method (Brent,

1973) is a bracketed root-finding algorithm that combines bisection with inverse quadratic interpolation, guaranteeing convergence within a user-specified bracket without requiring a derivative. At each trial m_s^{aq}, the full two-phase equilibrium is solved by an inner VLE solver (enforcing CO_2 and H_2O fugacity equality between phases), and the resulting phase amounts and compositions are used to evaluate the H_2O mass-balance residual; the Brent algorithm then bisects toward the m_s^{aq} at which this residual is zero. Initial guesses (warm starts) for the inner VLE solver are obtained from a molality-continuation cache built by marching from $m_s^{\text{aq}} = 0$ to an upper bound in small steps. This approach requires no initial guess for the full phase-state vector, making it highly robust. However, each function evaluation requires solving the eCPA phase compressibilities via nested Newton solves, making the outer loop expensive and ill-suited to the large number of evaluations required in a parameter-space scan or a production reservoir simulator. This motivated the development of the direct SSI approach described below.

3.2. Michelsen tangent-plane distance stability test

We implement the Michelsen tangent-plane distance (TPD) criterion (Michelsen, 1982) to determine whether a mixture at given $(T, P, \mathbf{z})$ is stable (single phase) or unstable (two-phase). Here $\mathbf{z} = (z_1, z_4)$ are the overall feed mole fractions of the two molecular species (NaCl is not an independent flash variable but enters through the molality-dependent fugacity coefficients, $\hat{\phi}_i$, below). Following Michelsen (1982), we introduce the vector of *unnormalized* trial mole numbers $\mathbf{W} = (W_1, W_4)$, which do not in general sum to unity. The normalized trial mole fractions are $w_i = W_i / \sum_j W_j$, and the TPD function is

$$\text{TPD}(\mathbf{W}) = \sum_i W_i \left[\ln W_i + \ln \hat{\phi}_i(\mathbf{w}) - d_i \right], \tag{9}$$

where the sum is over $i \in \{1, 4\}$, $d_i = \ln z_i + \ln \hat{\phi}_i(\mathbf{z})$ are reference chemical-potential offsets evaluated at the feed composition, and $\hat{\phi}_i$ is the fugacity coefficient. At any stationary point of TPD, one can show that TPD $= 1 - \sum_i W_i$, so a negative minimum (equivalently, $\sum_i W_i > 1$ at convergence of the SSI) indicates thermodynamic instability. Initial trial compositions are specified as normalized mole fractions derived from each K-value set (e.g. for the Wilson CO_2-rich trial, $w_1^{(0)} = K_1 z_1 / \sum_i K_i z_i$; for the fixed trials, $y_1 = 0.01$ or $x_1 \approx 0.999$).

Following Jex et al. (2024), we initialize the stability SSI from six trial K-value sets to ensure that no unstable stationary point is missed:

1. Wilson's correlation: $K_i^W = (P_{c,i}/P) \exp[5.373(1 + \omega_i)(1 - T_{c,i}/T)]$,
2. Inverse Wilson: $K_i = 1/K_i^W$,
3. Cube-root Wilson: $K_i = (K_i^W)^{1/3}$,
4. Inverse cube-root: $K_i = (1/K_i^W)^{1/3}$,
5. CO_2-rich: $K_{CO_2} = 0.95/z_{CO_2}$, $K_{H_2O} = 0.05/z_{H_2O}$,
6. H_2O-rich: $K_{H_2O} = (1 - 10^{-15})/z_{H_2O}$, $K_{CO_2} = 10^{-15}/z_{CO_2}$.

Each trial is iterated to convergence or a maximum of 200 SSI steps using the accelerated scheme described in Section 3.3. The mixture is classified as unstable if any trial converges to TPD $< -10^{-7}$; the K-values from the most negative TPD are passed to the flash as initial guess (Section 3.3).

As demonstrated in the parameter-space scan of Section 5.1, all six initial guesses play a material role: no single guess is best across the full (T, P, z) space, and all six are needed to ensure the optimal K-value initialization for the subsequent flash (Fig. 6b).

In practice, the stability test is the most computationally expensive step (~30 ms per call for the eCPA system), because it requires iterating the inner equations at each SSI step for each of the six trials. In the fast-flash algorithm described in Section 3.6, the stability test is bypassed for conditions pre-classified as two-phase by the solution table, reducing this overhead to zero for the vast majority of reservoir simulation cells.

EoS root consistency in the stability reference. The reference $d_i = \ln z_i + \ln \hat{\phi}_i(\mathbf{z})$ and all trial-phase evaluations use the same EoS, but the highly nonlinear CPA and eCPA equations of state can admit multiple real compressibility roots at a given condition, and the number of roots cannot be determined a priori. If the reference solver converges to a different root than the trial-phase solver, the resulting TPD comparison is between fugacities belonging to physically distinct solution branches, and the sign of TPD carries no thermodynamic meaning. This root-mixing artefact is most pronounced at pressures below the saturation pressure of water, $P < P_{\text{sat}}(H_2O, T)$: an H_2O-rich feed composition (large z_{H_2O}) can simultaneously support a liquid-like root ($Z \ll 1$) and a gas-like root ($Z \approx 1$), among others. A naive implementation that returns the *first* valid root found risks root-mixing: for H_2O-rich compositions at low pressure the reference may converge to the liquid root ($Z \ll 1$), while CO_2-rich trial iterates at small y_1 converge to the gas root ($Z \approx 1$), producing spuriously large negative TPD values and false two-phase predictions. Our implementation avoids this by collecting *all* distinct valid roots from multiple cold starts and selecting the one that minimizes the residual Gibbs energy $G^{\text{res}}/RT = \sum_i x_i \ln \hat{\phi}_i$. This is a purely thermodynamic criterion: the minimum-G root is the thermodynamically stable single-phase state at any $(T, P, \mathbf{x})$, independent of initial guess ordering or any pressure threshold. In the parameter-space scan (Section 5.1), this criterion eliminates nine false-positive two-phase predictions at $T = 190$–$220\,°\text{C}$, $P = 10\,\text{bar}$ (below P_{sat} for those temperatures), conditions at which no genuine two-phase equilibrium exists according to the EoS.

3.3. Successive-substitution isothermal flash

For two-phase conditions, we solve the equilibrium equations

$$x_i\, \hat{\phi}_i^{\text{aq}}(x, T, P) = y_i\, \hat{\phi}_i^{c}(y, T, P), \quad i = \text{H}_2\text{O},\ \text{CO}_2, \tag{10}$$

which are iterated by successive substitution on the K-values $K_i = y_i/x_i$. At each iteration k, the fugacity-ratio residual vector is

$$g_i^{(k)} = \ln \hat{\phi}_i^{c}(y^{(k)}) + \ln y_i^{(k)} - \ln \hat{\phi}_i^{\text{aq}}(x^{(k)}) - \ln x_i^{(k)}, \tag{11}$$

and the K-value update is

$$\ln K_i^{(k+1)} = \ln K_i^{(k)} - m^{(k)}\, g_i^{(k)}, \tag{12}$$

where $m^{(k)}$ is a step-size acceleration factor, discussed next.

Accelerated SSI. Following the dominant-eigenvalue acceleration of Jex et al. (2024), the step size is

$$m^{(k)} = \left| \frac{\mathbf{g}^{(k-1)\top}\, \mathbf{g}^{(k-1)}}{\mathbf{g}^{(k-1)\top}\left[\mathbf{g}^{(k-1)} - \mathbf{g}^{(k)}\right]} \cdot m^{(k-1)} \right|, \quad m^{(k)} \in [1,\ 1/L], \tag{13}$$

with $L = 0.1$ (so $m^{(k)} \leq 10$) for two-phase systems, and $m^{(1)} = 1$ (standard SSI on the first step). The formula estimates the dominant eigenvalue of the iteration map and extrapolates the K-value update along the slowest-decaying direction, dramatically reducing the number of iterations needed for convergence. Following Jex et al. (2024), we use the identity matrix as the scaling operator $\mathbf{U}^{-1}$ in the inner product, which is exact for two-component systems. The accelerated step $m^{(k)}g_i^{(k)}$ is clipped to $[-5, +5]$ in log-K space to prevent overshooting.

Convergence is declared when $\|\mathbf{g}^{(k)}\| < \tau$ with tolerance $\tau = 10^{-10}$.

Newton polish. Once the SSI residual falls below a switchover threshold τ_{switch} (default 10^{-4}), the solver switches to a Newton–Raphson step for quadratic convergence. At each Newton step, we solve the 2×2 linear system

$$\mathbf{J}^{(k)}\,\Delta \ln \mathbf{K}^{(k)} = -\mathbf{g}^{(k)}, \tag{14}$$

where the Jacobian $\mathbf{J} = \partial \mathbf{g}/\partial \ln \mathbf{K}$ is computed analytically. Each column of $\mathbf{J}$ requires the composition derivative $\partial \ln \hat{\phi}_i/\partial x_j$, which is obtained via the implicit function theorem applied to the three coupled inner equations for $(Z, \chi_{1w}, \chi_{1c})$: a 3×3 linear system is solved analytically at the converged SSI state, yielding $\mathrm{d}Z/\mathrm{d}x$, $\mathrm{d}\chi_{1w}/\mathrm{d}x$, and $\mathrm{d}\chi_{1c}/\mathrm{d}x$, from which $\partial \ln \hat{\phi}_i/\partial x_j$ follows by the chain rule. The Newton step is clipped to $[-5, +5]$ in $\ln K$ space. If Newton fails to reduce $\|\mathbf{g}\|$ or if the Jacobian is singular, the solver reverts to the pre-Newton state and continues with accelerated SSI, ensuring robustness. Across 9,825 two-phase conditions, Newton polish is triggered at 85.6% of points and reduces the mean iteration count from 9.8 (SSI only) to 6.9 (4.8 SSI + 2.1 Newton); see Table 4 and Fig. S10.

The CO_2-rich phase mole fraction β (the fraction of total moles in the CO_2-rich phase) is obtained at each step by solving the Rachford–Rice equation:

$$\sum_i \frac{z_i(K_i - 1)}{1 + \beta(K_i - 1)} = 0, \tag{15}$$

where the sum is over $i \in \{1, 4\}$ and z_i are the overall feed mole fractions. Initial K-values are taken from Wilson's correlation, from the stability test (Section 3.2), or from the solution table (Section 3.6.1).

Hierarchical flash algorithm. The stability test and flash are combined in a hierarchical algorithm (Jex et al., 2024): (1) run the TPD stability test with all six initial guesses; if stable, return single-phase; (2) use K-values from the most negative TPD trial as the initial guess for the accelerated SSI flash; (3) if the flash fails, fall back to Wilson K initialization. This strategy provides both robustness (100% convergence across all 9,825 two-phase conditions in the parameter-space scan, Section 5.1) and efficiency (stability-informed K-values reduce flash iterations by an additional 8% compared to Wilson initialization).

For cold-start SSI (without a table guess), a damping factor $\omega = 0.7$ is used in the standard (non-accelerated) variant to avoid divergence; warm-started SSI (from a table guess with state-vector error $\lesssim 0.01$) converges reliably with $\omega = 1.0$ in 3–4 iterations.

Inner Newton–Raphson solve for Z and χ (CPA). Within each SSI iteration, the CPA fugacity coefficients require simultaneous solution of the compressibility factor Z and the association site fractions χ for each phase. These quantities are mutually dependent: Z sets the molar volume and hence the packing fraction η, which enters the radial distribution function $g(\eta)$ and therefore the association strength Δ; the association fractions χ in turn feed back into the compressibility equation through the Wertheim term. For the CO_2–H_2O binary, the inner problem comprises three coupled equations per phase:

$$F_1 = Z - \frac{Z}{Z-B} + \frac{A}{Z+B} - \left(1 + \frac{\eta}{g}\frac{\mathrm{d}g}{\mathrm{d}\eta}\right)\left[x_w(\chi_w - 1) + x_c(\chi_c - 1)\right] = 0, \tag{16}$$

$$F_2 = c_3\chi_w^3 + c_2\chi_w^2 + c_1\chi_w + c_0 = 0, \tag{17}$$

$$F_3 = \chi_c\big(Z + 2x_w\chi_w\delta_1\big) - Z = 0, \tag{18}$$

where the coefficients c_0–c_3 in F_2 depend on Z through $\delta = g(\eta)\kappa\,[\exp(\varepsilon/k_BT) - 1]$, $\delta_1 = S_{cw}\,\delta$ is the scaled cross-association strength, and A, B are the dimensionless SRK parameters defined in Section 2. Equation (16) is the residual of the compressibility equation, (17) is the cubic equation for the H_2O bonding fraction χ_w (four sites, 4C scheme), and (18) is the algebraic closure for the CO_2 cross-association fraction χ_c. These three equations are solved simultaneously by Newton–Raphson, using the solution from the previous SSI iteration as the initial guess. The 3×3 Jacobian is derived analytically; all partial derivatives of F_1, F_2, and F_3 with respect to (Z, χ_w, χ_c) are available in closed form via the chain rule through $g(\eta)$ and its first two derivatives (Section S3.1). A scan+Brent fallback is retained for robustness when no warm start is available. The analogous but structurally different inner solve for the eCPA aqueous phase, which additionally involves the relative permittivity ε_r, is described in the next Section.

3.4. K-value flash for the eCPA ternary system

In the full eCPA ternary (CO_2 + H_2O + NaCl), the equilibrium NaCl molality in the aqueous phase m_s^{aq} couples to the flash equations through the aqueous-phase fugacity coefficients. Since NaCl is non-volatile, all salt remains in the aqueous phase and m_s^{aq} is fully determined by the current K-values and the feed composition — it carries no independent degree of freedom. We exploit this to *eliminate* m_s^{aq} analytically within each SSI step: given the current (K_1, K_4), the compositions of both phases and the molality are computed algebraically before the fugacity-coefficient evaluation, so the entire flash reduces to a single-level SSI on (K_1, K_4) with no outer loop.

Analytical elimination of m_s^{aq}. Given $K_1 \equiv y_1/x_1$ and $K_4 \equiv y_4/x_4$, the compositions of both phases are fixed once x_1 is known. On a one-mole volatile-feed basis with CO_2 feed fraction z_4 (so $z_1 = 1 - z_4$ is the H_2O feed fraction) and feed molality m_{feed}, the H_2O molar balance relates β to x_1: $\beta(x_1) = [z_1/x_1 - 1]\,/\,(K_1 - 1)$. The CO_2-rich phase normalization ($y_1 + y_4 = 1$, no NaCl) gives $x_4 = (1 - K_1\,x_1)/K_4$. Substituting $x_2 = x_3 = x_1\,m_s^{\mathrm{aq}}\,M_w$ and the NaCl molar balance $m_s^{\mathrm{aq}} = m_{\mathrm{feed}}\,z_1/[(1-\beta)\,x_1]$ into the aqueous normalization $\sum_i x_i = 1$ yields a single scalar equation in x_1:

$$x_1 + \frac{2\,m_{\mathrm{feed}}\,z_1\,M_w}{1 - \beta(x_1)} + \frac{1 - K_1\,x_1}{K_4} = 1. \tag{19}$$

Equation (19) is monotone in x_1 over the physical domain and is solved by Brent's method at each SSI step. Because this is a purely algebraic equation (no EoS evaluations), the scalar root-find adds negligible cost relative to the fugacity-coefficient evaluations that follow. Once x_1 is known, all compositions and the molality follow analytically:

$$m_s^{\mathrm{aq}} = \frac{m_{\mathrm{feed}}\,z_1}{[1 - \beta(x_1)]\,x_1}. \tag{20}$$

K-value iteration and acceleration. With compositions fully determined by the current (K_1, K_4), the fugacity residual is a 2-vector

$$g_i^{(k)} = \ln\hat{\phi}_i^{c}\big(y_1^{(k)}\big) + \ln y_i^{(k)} - \ln\hat{\phi}_i^{\mathrm{aq}}\big(x_1^{(k)}, m_s^{\mathrm{aq},(k)}\big) - \ln x_i^{(k)}, \quad i \in \{\mathrm{H_2O},\ \mathrm{CO_2}\}, \tag{21}$$

and the K-value update follows Eq. (12) with the Jex acceleration (Eq. (13)).

Inner Newton–Raphson solve for Z_w, ε_r, and χ_w (eCPA). Within each SSI iteration, the eCPA aqueous-phase fugacity coefficients require simultaneous solution of Z_w, the relative permittivity ε_r, and the H_2O association fraction χ_w. The permittivity is coupled to the association state through the Onsager–Kirkwood self-consistency equation (Maribo-Mogensen et al., 2015), and both feed back into the compressibility through the Debye–Hückel and permittivity correction terms absent in the salt-free CPA. The CO_2 cross-association fraction is eliminated algebraically, $\chi_c = Z_w/D_c$ with $D_c = Z_w + 2x_w\chi_w S_{cw}\Delta_w$, reducing the aqueous inner problem to three coupled equations:

$$G_0 = Z_w - \frac{Z_w}{Z_w - B_w} + \frac{A_w}{Z_w + B_w} - \left(1 + \frac{\eta}{g_w}\frac{\mathrm{d}g_w}{\mathrm{d}\eta}\right)\big[x_w(\chi_w - 1) + x_c(\chi_c - 1)\big] - Z_w^{\mathrm{DH}} - Z_w^{\mathrm{Perm}} = 0, \tag{22}$$

$$G_1 = \frac{(\varepsilon_r - \varepsilon_\infty)(2\varepsilon_r + \varepsilon_\infty)}{\varepsilon_r(\varepsilon_\infty + 2)^2} - \frac{N_A\rho\,\mu_{0w}^2}{9\varepsilon_0 k_B T}\,x_w\,g_w(\chi_w) = 0, \tag{23}$$

$$G_2 = \chi_w - \frac{Z_w}{Z_w + 2x_w\chi_w\Delta_w + 2x_c\chi_c S_{cw}\Delta_w} = 0. \tag{24}$$

Equation (22) is the eCPA compressibility residual: the SRK physical term is augmented by the Wertheim association correction (as in CPA) and by the Debye–Hückel compressibility Z_w^{DH} and the permittivity correction Z_w^{Perm} (Coelho et al., 2025), both of which depend on ε_r. Equation (23) is the Onsager–Kirkwood self-consistency condition: its left-hand side is the Onsager dielectric function of ε_r and $\varepsilon_\infty(Z_w)$ (the high-frequency permittivity from the Clausius–Mossotti relation), while the right-hand side couples the H_2O molar density ρ to the Kirkwood g-factor $g_w(\chi_w)$ through the permanent dipole moment μ_{0w} (Coelho et al., 2025). Equation (24) is the self-consistency condition for the H_2O bonding fraction, with $\Delta_w = g_w(\eta)\,\kappa_w[\exp(\varepsilon_w/k_BT) - 1]$ the association strength. These three equations are solved simultaneously by Newton–Raphson, warm-started from the previous SSI iterate. The 3×3 Jacobian $\mathbf{H} = \partial(G_0, G_1, G_2)/\partial(Z_w, \varepsilon_r, \chi_w)$ is derived analytically in Section S3.2. The CO_2-rich phase retains the 3×3 system of Section 3.3 (with $x_{1c} = 0$ eliminating the eCPA terms); when no warm start is available, a general-purpose root-finder provides the cold-start fallback for the aqueous phase. Convergence is declared when $\|\mathbf{G}^{(k)}\| < 10^{-8}$.

In the hierarchical algorithm described in Section 3.6, K-values are always initialized from the stability test result, with Wilson's correlation as a fallback. When the flash is invoked without a prior stability result, four fixed (K_1, K_4) seeds are tried in sequence until one converges.

Outer Newton polish for the eCPA K-value flash. By analogy with the CPA flash ("Newton polish" paragraph in Section 3.3), we test an outer Newton–Raphson polish applied once the K-value SSI residual $\|\mathbf{g}\|$ falls below a switchover threshold. Because the eCPA K-value residual couples the highly nonlinear electrostatic and association contributions through the implicitly defined aqueous composition x_1 (Eq. (19)), deriving an analytical 2×2 Jacobian $\partial\mathbf{g}/\partial \ln \mathbf{K}$ would require implicit differentiation through a scalar Brent solve at every Newton step. We therefore test the approach with a forward-difference numerical Jacobian, which adds two additional $\ln\phi$ evaluations per Newton step. The benchmarks (672 conditions spanning $T = 37$–$157\,°\mathrm{C}$, $P = 60$–$400\,\mathrm{bar}$, $z_{CO_2} = 0.15$–0.70, $m_s = 0.5$–$4.5\,\mathrm{mol\,kg^{-1}}$) show that on the warm-start path — the production path in sequential simulation — the SSI already converges in approximately one iteration because the table initial guess is close to the solution, so Newton is never triggered and the overhead is negligible. On the cold-start path, where SSI requires $\sim$7 iterations on average, the Newton polish reduces the total iteration count by 25% (from 7.15 to 5.35 for a switchover threshold of 10^{-3}, or 17% for 10^{-4}), but the numerical-Jacobian overhead — three $\ln\phi$ evaluations per

Newton step versus one for SSI — cancels the iteration savings in wall time ($\lesssim$ 1% improvement). Because the warm-start path dominates in practice and a worthwhile wall-time speedup would require an analytical Jacobian that is non-trivial to derive, the outer Newton polish is not applied by default in the eCPA flash.

3.5. Phase envelope

We trace the two-phase boundary (P–T locus at fixed z_4 and m_s) using the following procedure:

1. Scan a coarse grid of (T, P) points with the stability test to classify each cell as stable or unstable.
2. Identify transitions between stable and unstable cells.
3. Refine each transition to the phase boundary by bisection, using SSI to solve the flash at each trial point.

To handle the large number of (T, P) combinations efficiently, the bisection at each boundary point is executed in parallel across all available CPU cores.

3.6. Fast flash for reservoir simulation

3.6.1. Three-dimensional solution table

We precompute flash solutions on a three-dimensional grid (T, P, m_s) with:

- T: 361 points from 0 to 360 °C (1 °C spacing), covering conditions from cold-basin CO_2 storage to hydrothermal and geothermal applications.
- $\log_{10} P$: 100 points from $\log_{10}(1)$ to $\log_{10}(2000)$ (log-uniform).
- m_s (NaCl molality): 14 points from 0.0 to 6.0 mol kg^{-1}, spanning the full range of experimental data from dilute brine to near-saturation.

This gives 361 × 100 × 14 = 505,400 grid cells. Here m_s denotes the NaCl feed molality, i.e. moles of NaCl per kilogram of H_2O in the feed mixture; it differs from the equilibrium aqueous molality m_s^{aq}, which is larger due to the salting-out effect. The overall CO_2 feed fraction z_4 is not included as a table dimension. All table entries were computed at a representative feed fraction $z_4 = 0.5$; the equilibrium quantities stored (K-values, Z, χ, m_s^{aq}) depend weakly on z_4 for fixed (T, P, m_s). This dependence is primarily through the salting-out effect: a larger CO_2 feed fraction partitions more water into the CO_2-rich phase, concentrating the brine and raising m_s^{aq}, which in turn slightly shifts the K-values. We verified numerically that at typical subsurface conditions ($T = 50$ °C, $P = 100$ bar, $m_s = 1$ mol kg^{-1}) the spread of K-values across the full two-phase z_4 range is less than 0.15%, confirming that the table provides adequate warm-start quality across the full feed composition range. Excluding z_4 reduces the table size by more than an order of magnitude relative to a four-dimensional grid.

Each cell is solved using the full hierarchical algorithm (Section 3.2): the Michelsen stability test (six trial compositions) is performed first; only genuinely unstable conditions proceed to K-value SSI flash. This means no external assumption about the phase state is required; every cell is self-consistently classified and solved. At each two-phase cell we store the converged K-values (K_1, K_4), the inner-Newton state for the aqueous phase ($Z_w, \varepsilon_r, \chi_{1w}$) and for the CO_2-rich phase (Z_c, χ_{1c}), and the equilibrium aqueous molality m_s^{aq}. All seven quantities serve directly as warm-start inputs to the hierarchical flash: K-values initialize the outer SSI, while Z, ε_r, and χ initialize the inner Newton solves for compressibility factor and association fractions (Section S3.2), eliminating the need for cold-start Newton iterations. A scalar stability flag (two-phase or single-phase) is also stored.

Table lookup uses a trilinear interpolant operating in $(T, \log_{10} P, m_s)$ space, which is vectorized and evaluates in <1 ms.

3.6.2. Fast flash algorithm

The fast flash routine operates as follows. A query to the 3D interpolant in $(T, \log_{10} P, m_s)$ yields an initial state vector (compressibility factors, phase compositions, association fractions, and permittivity) together with a stability classification (two-phase or single-phase). If the interpolant classifies the condition as two-phase, the Michelsen stability test is skipped and SSI is initialized directly from the table state vector with no damping ($\omega = 1.0$). If the classification is single-phase or uncertain (for example, near the two-phase boundary or for grid cells with low convergence quality), the full Michelsen TPD test is performed; a stable result returns immediately without a flash, while an unstable result proceeds to SSI. In either two-phase path, the warm-started SSI converges in 3–4 iterations because the interpolated state vector is already within ∼1% of the converged solution. Should SSI fail to converge, a full cold-start with Wilson K-value initialization and damped SSI ($\omega = 0.7$) provides a robust fallback.

The key insight enabling this strategy is that the solution surface — the mapping from (T, P, m_s) to the equilibrium state vector — is smooth and slowly varying except near the two-phase boundary and the CO_2 critical point ($T_{c,c} = 31.3\,°\mathrm{C}$, $P_c = 73.8$ bar). A trilinear interpolant from a grid with 1 °C temperature spacing and log-uniform pressure spacing (100 points over 1–2000 bar) therefore provides an excellent initial guess for the vast majority of reservoir conditions. The selective stability check further reduces cost: in a typical reservoir simulation, the majority of grid cells are well within either a single-phase or a two-phase region (injected CO_2 coexisting with brine) and the interpolant stability flag can be trusted, while the expensive Michelsen test is reserved for cells near boundaries where phase transitions occur.

3.7. Alternative simplified flash exploiting $y_{H_2O} \approx 0$

At typical CO_2 storage reservoir conditions ($T \lesssim 100\,°\mathrm{C}$), the H_2O content of the CO_2-rich phase remains below 1 mol% (see Fig. 8 and the discussion in Section 5.1.2). The approximation $y_{\mathrm{H_2O}} = 0$ then collapses the standard two-component K-value flash to a single scalar equation, enabling a simpler, self-contained calculation without the outer salinity iteration.

Mathematical basis. Under $y_{\mathrm{H_2O}} = 0$ the CO_2-rich phase is pure CO_2, so its fugacity reduces to the pure-component value:

$$f^{\mathrm{c}}_{\mathrm{CO_2}}(T, P) = P\,\phi^{\mathrm{pure}}_{\mathrm{CO_2}}(T, P). \tag{25}$$

A compact material-balance argument shows that the aqueous NaCl molality equals the feed value exactly under this assumption. Define the effective H_2O feed mole fraction $z_1 = (1 - z_{\mathrm{CO_2}})/(1 + 2m_{\mathrm{feed}}M_w)$, where $M_w = 0.018015\,\mathrm{kg\,mol^{-1}}$. From the NaCl and H_2O balances (salt is non-volatile, so $z_{\mathrm{NaCl}} = (1-\beta)\,x_1^{\mathrm{aq}}\,m_s^{\mathrm{aq}}\,M_w$ and $z_1 = (1-\beta)\,x_1^{\mathrm{aq}}$), dividing gives $m_s^{\mathrm{aq}} = z_{\mathrm{NaCl}}/(z_1 M_w) = m_{\mathrm{feed}}$. The equilibrium is therefore determined by a single equation in x_1^{aq}:

$$f^{\mathrm{aq}}_{\mathrm{CO_2}}(x_1^{\mathrm{aq}},\, m_{\mathrm{feed}},\, T,\, P) \;=\; f^{\mathrm{c}}_{\mathrm{CO_2}}(T, P), \tag{26}$$

where $x_4^{\mathrm{aq}} = 1 - x_1^{\mathrm{aq}}(1 + 2m_{\mathrm{feed}}M_w)$ and the aqueous fugacity is evaluated with the eCPA 3×3 Newton solver (Section 3.4). The phase fraction follows from the H_2O material balance:

$$\beta = 1 - \frac{z_1}{x_1^{\mathrm{aq}}}. \tag{27}$$

Phase identification is embedded in the root-finding step: if the CO_2-fugacity residual evaluated at $x_1^{aq} = z_1$ (i.e. $\beta = 0$) is non-positive, the mixture is single-phase aqueous and no further solve is needed.

Algorithm. The pure-CO_2 fugacity (25) is computed once via the CO_2-rich-phase inner Newton solver at $x_1^c = 0$. Equation (26) is then solved by Brent's method over the bracket $[z_1, x_1^{\max}]$ where $x_1^{\max} = 1/(1 + 2m_{\mathrm{feed}}M_w)$ is the H_2O mole fraction of a salt-only aqueous phase. Warm-starting of the inner Newton solver across Brent evaluations (carrying forward $[Z_w, \varepsilon_r, \chi_{1w}]$ from each accepted step) reduces the inner-solve cost to the same 3–6 Newton iterations achieved in the full flash. The formulation is valid for both the salt-free CPA binary ($m_{\mathrm{feed}} = 0$) and the full eCPA ternary; no outer salinity iteration is required in either case. Results are assessed against the full K-value flash in Section 5.3.

4. Validation

4.1. CO_2 + H_2O binary

We validate both the (new and tuned) salt-free CPA EOS and the eCPA EOS at $m_s = 10^{-4}\,\mathrm{mol\,kg^{-1}}$ (effectively salt-free) against 631 experimental VLE data points compiled by Coelho et al. (2025) for $T = 0$–$350\,°\mathrm{C}$, achieving 100% convergence across all conditions. Both phases are characterised: the CO_2 mole fraction in the aqueous phase x_{CO_2} and the H_2O mole fraction in the CO_2-rich phase y_{H_2O}.

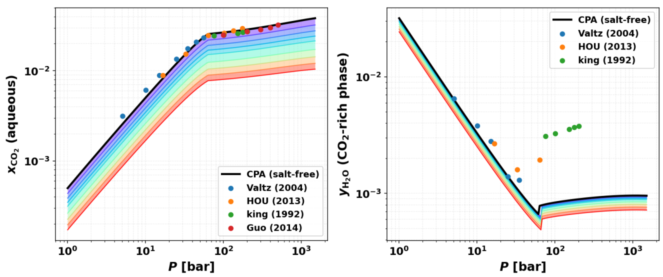
Fig. 1 shows representative comparisons at six temperatures spanning the validated range ($T = 25$–$250\,°\mathrm{C}$); results for all 35 temperatures are collected in Fig. S1 in the Supplemental Information (phase split results for molalities up to $m_s = 6\,\mathrm{mol\,kg^{-1}}$ are also included for comparison and discussed further below). Table 1 summarizes the accuracy metrics by condition regime. CPA and eCPA give essentially identical results for the salt-free binary (an encouraging consistency validation between independent codes), so only eCPA values are reported. The results in Fig. 1 use a feed composition of 50 mol% CO_2 and the full stability+flash framework; Coelho et al. (2025) report VLE compositions only (phase fractions are not needed for their parameterization validation).

Over all 631 conditions, the overall AARE is 9.6% for x_{CO_2} ($N = 560$) and 24.4% for y_{H_2O} ($N = 452$); the larger error for the CO_2-rich phase is consistent with the Coelho et al. (2025) parametrization, which prioritizes brine conditions. However, these headline numbers are inflated by three challenging regimes:

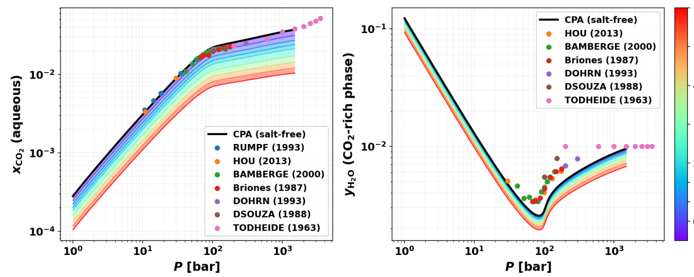
1. *Near the CO_2 critical point* ($T_{c,c} = 31.3\,°\mathrm{C}$, $P_c = 73.8\,\mathrm{bar}$): K-value gradients with pressure are steep and small errors in fugacity coefficients translate to large errors in solubility, consistent with the general difficulty of describing near-critical behavior with equations of state based on a cubic repulsive term.
2. *Near the H_2O critical point* ($T \geq 260\,°\mathrm{C}$): the two-phase envelope narrows and the mixture critical locus predicted by the EoS deviates slightly from experiment. At $T = 268$–$270\,°\mathrm{C}$, experimental compositions become identical ($K_i = 1$), indicating conditions at or beyond the mixture critical point.
3. *Ultra-high pressures* ($P > 1500\,\mathrm{bar}$): beyond typical reservoir depths and outside the solution-table range. The sole data source at $P > 2000\,\mathrm{bar}$ (Tödheide and Franck, 1963) cannot be independently cross-validated, and errors in this regime reach 78% AARE for y_{H_2O}.

For conditions most relevant to subsurface CO_2 storage ($T = 30$–$150\,°\mathrm{C}$, $P = 50$–$600\,\mathrm{bar}$), the AARE improves to **6.5%** for x_{CO_2} ($N = 137$).

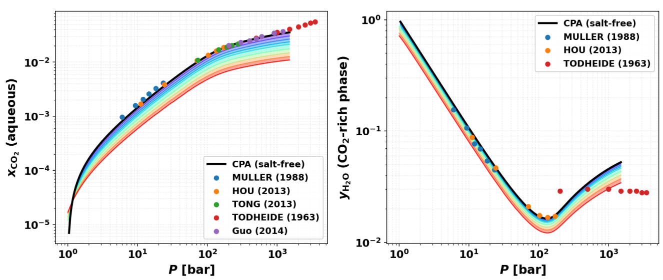
A cross-reference consistency analysis (Fig. S7) further reveals that the y_{H_2O} data of King et al. (1992) are physically suspect: the reported water content in the CO_2-rich phase *increases* with pressure above the CO_2 saturation pressure, contradicting thermodynamic expectations and the measurements of Meyer and Harvey (2015) at overlapping conditions. After excluding flagged outlier references (92 of 631 points, see Fig. S7), the overall y_{H_2O} AARE decreases from 24.4% to 17.4%.



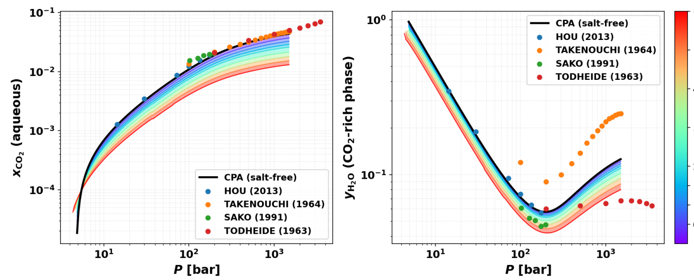
(a) 25 °C (298 K)



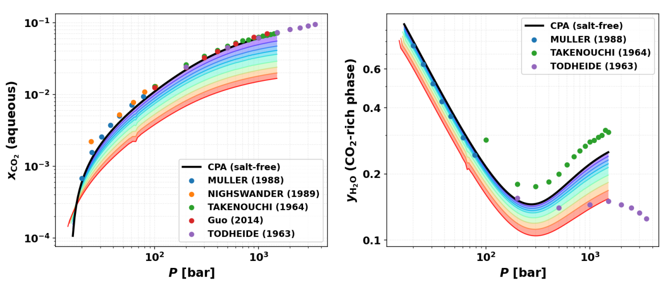
(b) 50 °C (323 K)



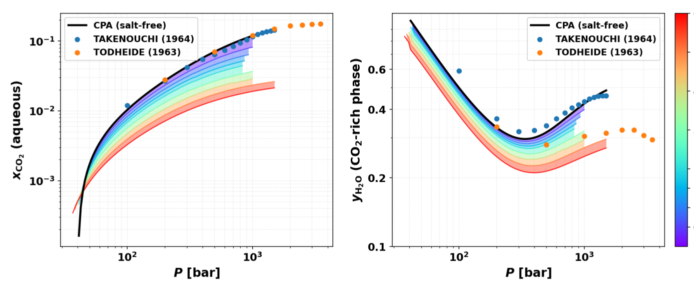
(c) 100 °C (373 K)



(d) 150 °C (423 K)



(e) 200 °C (473 K)



(f) 250 °C (523 K)

Figure 1: $CO_2 + H_2O$ phase equilibria at six representative temperatures (25–250 °C). For each temperature, the left panel shows the CO_2 mole fraction in the aqueous phase (x_{CO_2}) and the right panel shows the H_2O mole fraction in the CO_2-rich phase (y_{H_2O}) vs. pressure. Open symbols: experimental data (Coelho et al., 2025); black solid line: CPA (salt-free); rainbow-colored filled bands: eCPA predictions spanning NaCl molalities $m_s = 0$–$6\,\mathrm{mol\,kg^{-1}}$ (filled between consecutive molality contours; violet ≈ 0, red $= 6\,\mathrm{mol\,kg^{-1}}$), with discrete eCPA curves overlaid. Results for all 35 temperatures are shown in Fig. S1.

These model errors should be interpreted in the context of the experimental uncertainty in the compiled dataset. Two independent estimates of measurement scatter can be obtained directly from the data. First, at the 76 (T, P) conditions where at least two independent references overlap (matched within ± 2.5 bar), the coefficient of variation between reported values averages 4.0% for x_{CO_2} and 11.1% for y_{H_2O}. Second, the Gasser–Sroka–Jennen-Steinmetz (GSJS) difference-based variance estimator (Gasser et al., 1986), which estimates measurement noise from the scatter of consecutive data points around a locally interpolated trend within each isothermal series, yields a median relative noise of 2.4% for x_{CO_2} and 4.7% for y_{H_2O}. Taken together, these estimates suggest an intrinsic experimental uncertainty of roughly 2–4% for x_{CO_2} and 5–11% for y_{H_2O}, so the model AARE of 6.5% (subsurface) to 9.6% (global) for x_{CO_2} are close to the measurement noise.

Moreover, relative errors can be misleading when mole fractions span several orders of magnitude: a 35% AARE for $y_{H_2O} < 0.005$ (101 points) corresponds to an average absolute error (AAE) of only 0.1 mol%—negligible for reservoir simulation. Across all data, the AAE is 0.53 mol% for x_{CO_2} and 3.5 mol% for y_{H_2O}; under subsurface conditions these drop to 0.13 and 0.68 mol%, respectively.

Table 1: VLE accuracy metrics for the CO_2 + H_2O binary by condition regime. AARE: average absolute relative error; AAE: average absolute error (mol%). CPA and eCPA ($m_s = 10^{-4}$) give identical results for this salt-free system; only eCPA values are shown. "Excl. outliers" removes references flagged by cross-reference consistency (see text).

	x_{CO_2} (aqueous)			y_{H_2O} (CO_2-rich)		
Regime	N	AARE	AAE	N	AARE	AAE
All data	560	9.6%	0.53 mol%	452	24.4%	3.5 mol%
All (excl. outliers)	468	9.3%	0.53 mol%	355	17.4%	3.0 mol%
Subsurface	137	6.5%	0.13 mol%	102	25.6%	0.68 mol%
Extended (T = 10–200 °C)	362	8.9%	0.21 mol%	299	24.6%	1.7 mol%
High-T ($T \geq 260$ °C)	129	10.1%	1.6 mol%	111	15.9%	8.0 mol%
High-P ($P > 1500$ bar)	27	18.4%	2.9 mol%	27	78.0%	15.0 mol%

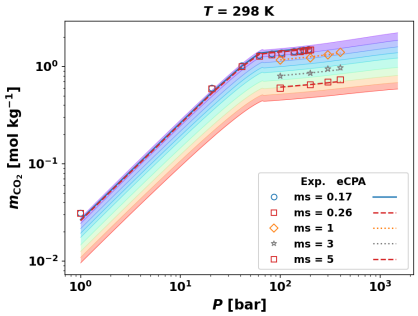
Subsurface: T = 30–150 °C, P = 50–600 bar.

4.2. CO_2 + NaCl brine

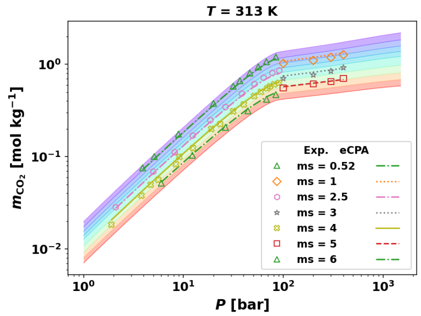
Fig. 1 also illustrates the effect of brine salinity on the equilibrium compositions by including predictions for the full molality range considered in this work. We further validate the eCPA ternary flash against all available experimental data from the literature (Guo et al., 2015; Rumpf et al., 1994; Koschel et al., 2006; Kiepe et al., 2002; Takenouchi and Kennedy, 1965; Chabab et al., 2020; Messabeb et al., 2016; Hou et al., 2013; Yan et al., 2011; Zhao et al., 2015; Bando et al., 2003; Nighswander et al., 1989; Liu et al., 2021, 2011; Savary et al., 2012), spanning T = 5–450 °C, P = 1–1400 bar, and m_s = 0.17–6.0 mol kg^{-1} NaCl (708 experimental data points in total). Two types of measurements are compared: CO_2 solubility in the aqueous phase and the CO_2 mole fraction in the CO_2-rich phase (y_{CO_2}). Aqueous-phase solubility is reported differently by different sources: as CO_2 molality (m_c [mol kg^{-1}]), or as the mole fraction on a NaCl-free basis ($x^{sf}_{CO_2} = x_{CO_2}/(x_{CO_2} + x_{H_2O})$), where NaCl is excluded from the denominator. The latter is a reporting convention used by some authors and describes the same physical quantity as m_c; it is converted to molality via $m_c = x^{sf}_{CO_2}/[(1 - x^{sf}_{CO_2})M_w]$ before comparison.

Table 2: VLE accuracy metrics for CO_2 + NaCl brine (T = 5–450 °C, m_s = 0.17–6.0 mol kg^{-1}, 708 data points total, 689 two-phase converged). Bias = $\langle(\hat{y} - y)/y\rangle$.

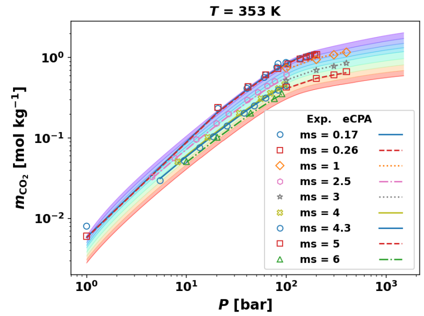
Quantity	N	AARE (%)	Bias (%)
m_c [mol kg^{-1}]	501	6.6	−3.3
$x^{sf}_{CO_2}$ (NaCl-free basis, conv. to m_c)	116	5.5	+2.7
x_{CO_2} [salt-incl.]	36	8.6	+8.6
y_{CO_2}	36	0.5	−0.1
All converged	689	6.2	−1.5



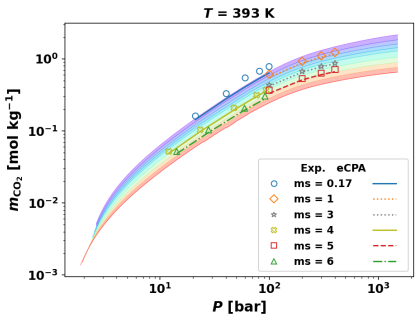
(a) 25 °C



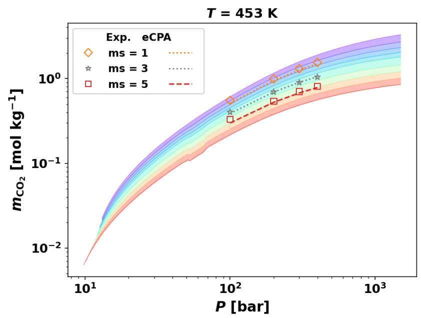
(b) 40 °C



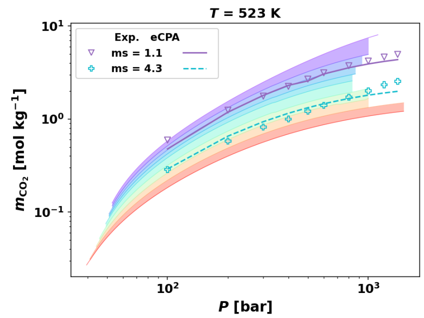
(c) 80 °C



(d) 120 °C



(e) 180 °C



(f) 250 °C

Figure 2: CO_2 solubility in NaCl brine at six representative temperatures (25–250 °C). Each panel shows CO_2 molality m_c [mol kg^{-1}] vs. pressure (log–log). Open symbols: experimental data (Guo et al., 2015; Rumpf et al., 1994; Kiepe et al., 2002; Nighswander et al., 1989; Bando et al., 2003; Liu et al., 2021; Takenouchi and Kennedy, 1965); discrete lines with markers: eCPA at the experimental NaCl molalities. Rainbow-colored background bands: continuous eCPA predictions spanning m_s = 0–6 mol kg^{-1} (color scale as in Fig. 1: violet ≈ 0, red = 6 mol kg^{-1}). Temperatures with additional CO_2-rich phase data (y_{CO_2}) are shown separately in Fig. 3; all 18 temperatures in Figs. S2 – S3.

All experimental results and eCPA predictions are illustrated in Figs. 2, 3, S2, and S3.

Table 2 shows the accuracy breakdown. 689 of 708 points converge to a two-phase solution using the K-value SSI warm-started from the solution table; the model predicts single-phase behavior at 18 points (particularly at $T \geq 350$ °C where $CO_2 + H_2O$ approaches complete miscibility in the model) and fails to converge at one extreme high-P, high-m_s condition, giving an overall two-phase convergence rate of 97.3%. The K-value SSI requires a median of 6 outer iterations per converged call. Overall AARE across all converged points is 6.2%, with a slight underprediction bias of −1.5%. The AARE for m_c across all T and m_s is 6.6%, in excellent agreement with the 6.9% reported by Coelho et al. (2025) using the original eCPA parameterization. The CO_2-rich phase composition y_{CO_2} is reproduced to within 0.4% AARE, reflecting the near-purity of the CO_2-rich phase and its weak dependence on salinity.

At hydrothermal temperatures, AARE for m_c rises to 14.0% at $T = 300$ °C and 12.8% at $T = 350$ °C, reflecting reduced eCPA accuracy at conditions beyond the conventional reservoir temperature range. At $T \geq 400$ °C, the available experimental pressures (300–900 bar, all from Takenouchi and Kennedy 1965) fall in the region where eCPA predicts single-phase behavior, suggesting that the $CO_2 + H_2O$ miscibility gap closes at these extreme conditions in the model. Notably, accuracy at the lowest temperatures (T = 5–10 °C) is also reduced (AARE 14–21%), reflecting the challenge of modeling CO_2 solubility very close to the freezing point where eCPA is not specifically parameterized. Fig. S4 (SI) provides a comprehensive view of the error distribution across the full (T, m_s) parameter space: accuracy is best at intermediate temperatures (T = 27–227 °C) and moderate salinities ($m_s \leq 3$ mol kg^{-1}), and degrades systematically at the boundaries of the validated range.

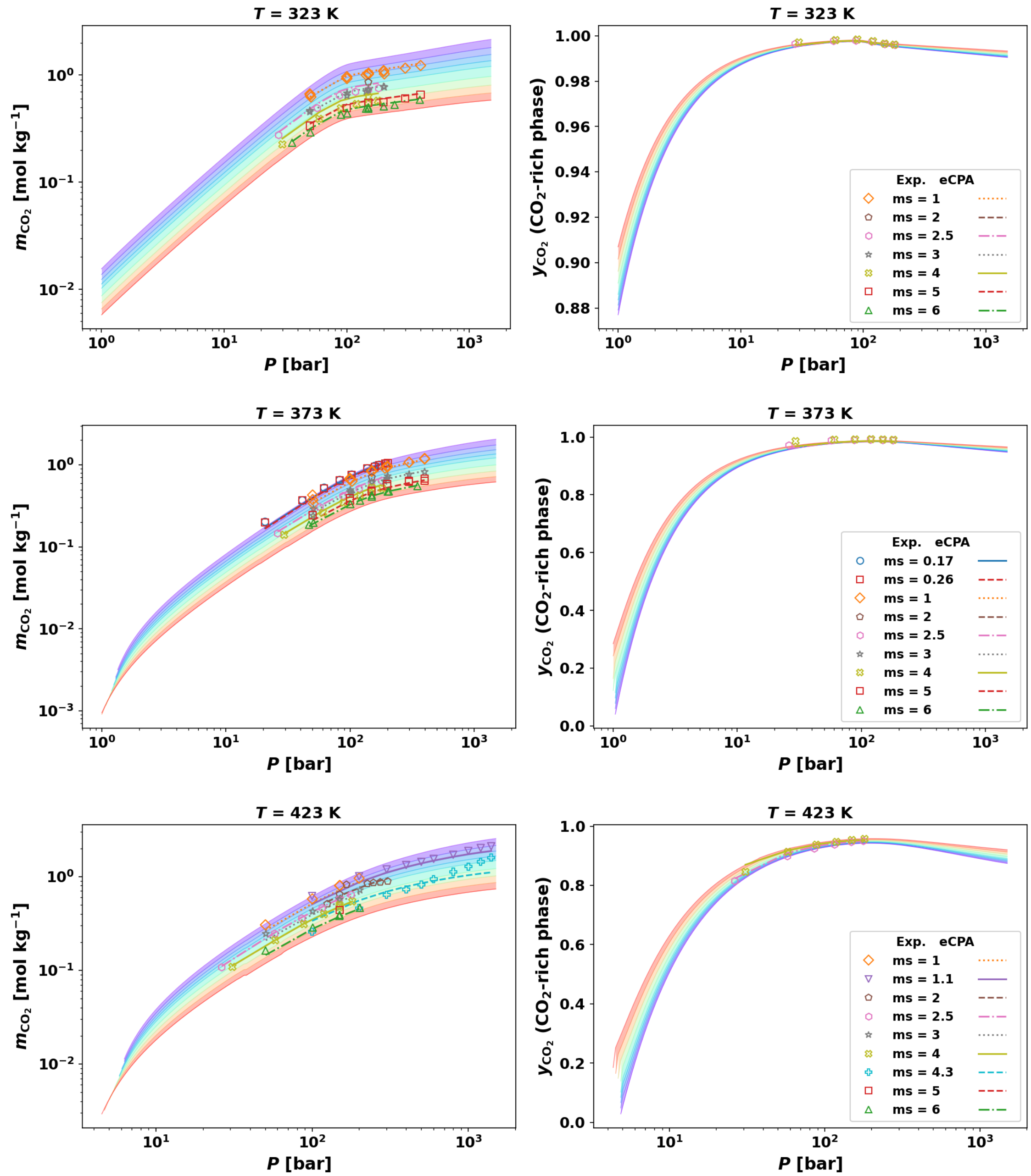


Figure 3: CO_2 + NaCl brine phase equilibria at temperatures for which both CO_2 molality (m_c, left panel) and CO_2 mole fraction in the CO_2-rich phase (y_{CO_2}, right panel) data are available: (top) 50 °C, (middle) 100 °C, (bottom) 150 °C. Open symbols: experimental data (Bando et al., 2003; Chabab et al., 2020; Guo et al., 2015; Hou et al., 2013; Koschel et al., 2006; Liu et al., 2021; Messabeb et al., 2016; Savary et al., 2012; Takenouchi and Kennedy, 1965; Yan et al., 2011; Zhao et al., 2015); discrete lines with markers: eCPA at the experimental NaCl molalities. Rainbow-colored background bands: continuous eCPA predictions across $m_s = 0$–$6\,\mathrm{mol\,kg^{-1}}$ (color scale as in Fig. 1).

4.3. Flash results as a function of feed composition

A central motivation for full stability+flash — rather than straightforward VLE calculations at a fixed salinity — arises naturally in reservoir simulation. When CO_2 is injected into a saline aquifer, the reservoir fluid at any given grid cell can be represented as a mixture of brine (carrying NaCl at feed molality m_{feed}) and varying amounts of injected CO_2. The overall CO_2 mole fraction z_4 in this mixture is the primary degree of freedom, and stability+flash must be performed to determine the equilibrium state.

As shown in Section 1, in the ternary system the equilibrium molality m_s^{aq} exceeds the feed value because NaCl is concentrated as H_2O partitions into the CO_2-rich phase; quantitatively (from Eq. 20):

$$m_s^{\mathrm{aq}} = m_{\mathrm{feed}} \frac{(1-\beta)\, x_{\mathrm{H_2O}} + \beta\, y_{\mathrm{H_2O}}}{(1-\beta)\, x_{\mathrm{H_2O}}} = m_{\mathrm{feed}} \left(1 + \frac{\beta\, y_{\mathrm{H_2O}}}{(1-\beta)\, x_{\mathrm{H_2O}}}\right),$$

where β is the CO_2-rich molar phase fraction. Because $\beta\, y_{\mathrm{H_2O}} > 0$ in the two-phase region, m_s^{aq} *exceeds* m_{feed}, and the excess grows with z_4 as β increases. This elevated salinity modifies the aqueous-phase fugacity coefficients and in turn shifts the equilibrium phase compositions — an effect that cannot be captured by a VLE table indexed on feed molality alone.

Fig. 4 demonstrates these effects for $P = 200\,\mathrm{bar}$, $m_{\mathrm{feed}} = 1.0\,\mathrm{mol\,kg^{-1}}$ NaCl, and ten isotherms from 27 to 252 °C (in 25 °C increments) over the two-phase feed range $z_4 = 0.005$–0.995. Panel (a) shows the CO_2-rich phase fraction β, which rises from 0 to 1 across the two-phase window. Higher temperatures narrow the window because CO_2 is less soluble and H_2O more volatile, both effects shrinking the range of z_4 that sustains two-phase coexistence.

Panel (b) illustrates the salinity amplification: at the highest isotherm ($T = 252\,°\mathrm{C}$) the ratio $m_s^{\mathrm{aq}}/m_{\mathrm{feed}}$ reaches ≈ 8 at the high-z_4 end of the two-phase window, meaning the aqueous NaCl concentration increases eightfold relative to the injected brine as H_2O partitions into the growing CO_2-rich phase. At lower temperatures the amplification is smaller because less H_2O leaves the aqueous phase at the lower β values attained.

Panel (c) shows the resulting CO_2 molality in the aqueous phase, m_c [$\mathrm{mol\,kg^{-1}}$]. Despite operating at fixed (T, P), m_c decreases substantially with increasing z_4: at $T = 227\,°\mathrm{C}$ it falls from ≈ 1.1 to $\approx 0.4\,\mathrm{mol\,kg^{-1}}$ across the two-phase window. This is a direct consequence of the salting-out effect — the elevated m_s^{aq} suppresses CO_2 solubility. A simple VLE lookup at the feed molality m_{feed} would return the wrong CO_2 solubility for all but the dilute-CO_2 end of the window.

Panel (d) shows the H_2O mole fraction in the CO_2-rich phase, $y_{\mathrm{H_2O}}$, and reveals a second consequence of the salting-out chain. As z_4 increases, β increases and m_s^{aq} rises (panel b), which suppresses CO_2 solubility in the aqueous phase (panel c): the aqueous phase can dissolve less CO_2, so proportionally more CO_2 remains in the CO_2-rich phase and, by molar conservation, $y_{\mathrm{H_2O}}$ is diluted downward. At low temperatures this effect is invisible because the baseline water content of the CO_2-rich phase is tiny; at high temperatures, where $y_{\mathrm{H_2O}}$ starts near $\approx 35\,\mathrm{mol\%}$, the same salting-out suppression translates into a large absolute drop — falling to $\approx 22\,\mathrm{mol\%}$ by $z_4 \approx 0.75$ at $T = 252\,°\mathrm{C}$. This also underscores the importance of accurate modeling of the CO_2-rich phase at elevated temperatures.

Taken together, these results demonstrate that the correct representation of CO_2+brine phase behavior in reservoir models requires a full stability+flash computation for each cell at each time step, with the NaCl molality updated self-consistently from the aqueous-phase amount.

4.4. Aqueous-phase density

The CPA energy parameter and co-volume for H_2O are tuned to pure-water vapor pressure and liquid density, so the model is expected to reproduce aqueous densities without requiring a Péneloux volume shift. Both CPA and eCPA predict CO_2-saturated aqueous densities with **0.76%** AARE over 37

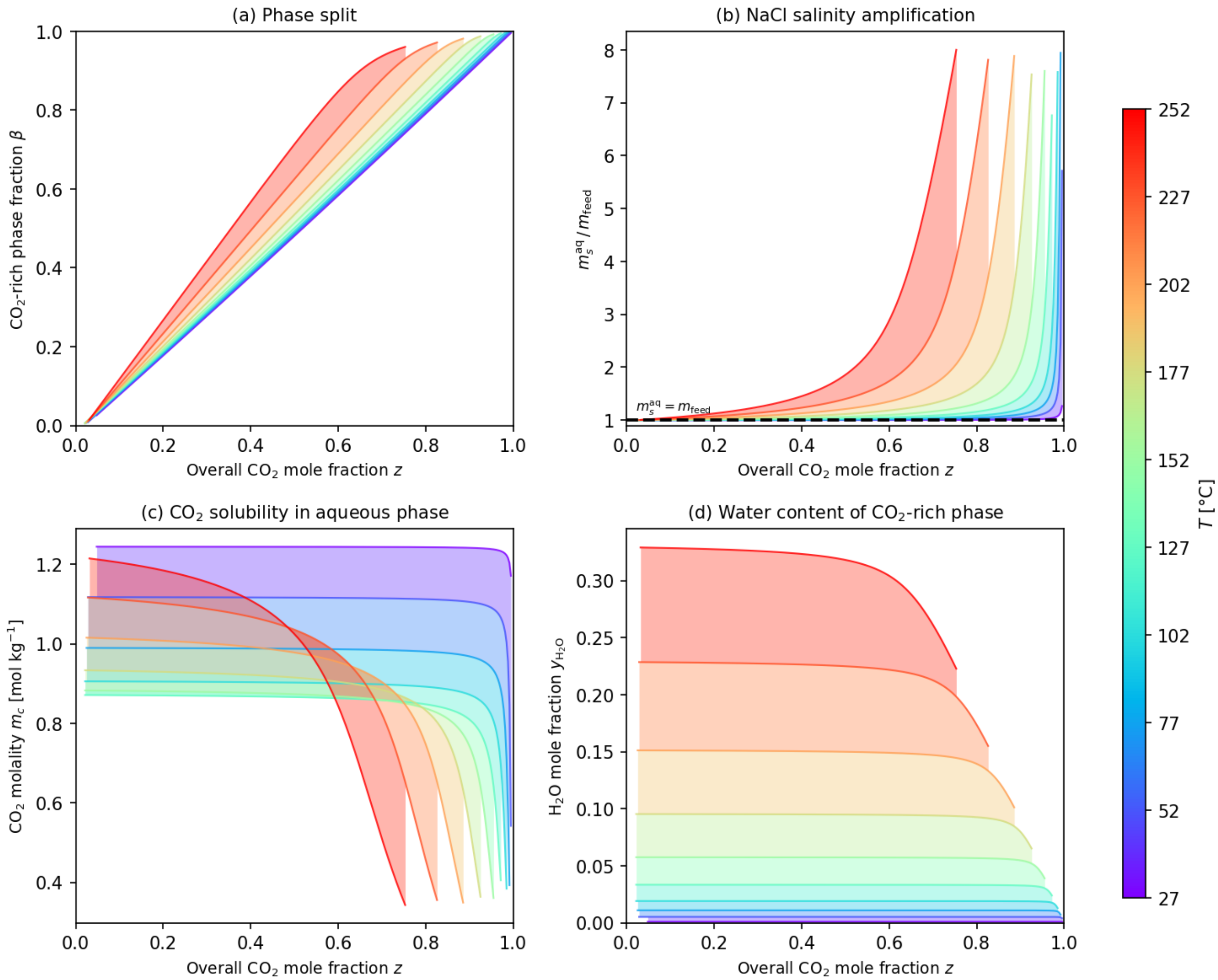


Figure 4: eCPA flash results as a function of overall CO_2 feed mole fraction z_4 at $P = 200\,\text{bar}$, feed molality $m_{\text{feed}} = 1.0\,\text{mol}\,\text{kg}^{-1}$ NaCl, and ten isotherms from 27 to 252 °C (violet to red). (a) CO_2-rich phase fraction β; the two-phase window narrows progressively at higher temperature. (b) Ratio of equilibrium aqueous NaCl molality to feed molality, $m_s^{\text{aq}}/m_{\text{feed}}$; NaCl concentrates in the aqueous phase as H_2O partitions to the CO_2-rich phase, reaching a factor of ∼8 at $T = 252\,°\text{C}$ and $z_4 \approx 0.98$. (c) CO_2 molality m_c in the aqueous phase; the salting-out effect driven by the elevated m_s^{aq} reduces CO_2 solubility with increasing z_4. (d) H_2O mole fraction y_{H_2O} in the CO_2-rich phase; substantial water content at high temperatures (up to ∼30 mol% at 252 °C) and its strong variation with z_4 highlight the need for full flash rather than a fixed-salinity VLE lookup.

experimental points ($T = 15$–$200\,°$C) from King et al. (1992) and Nighswander et al. (1989), confirming that no volume shift is needed.

To test density accuracy over the full (T, P) range of the paper rather than the limited set of available CO_2-saturation measurements, we compare pure-water eCPA predictions against the IAPWS-95 reference equation (Wagner and Pruß, 2002) — the internationally accepted scientific formulation for water properties, valid to 1 000 MPa and 1 273 K, and the underlying standard of the NIST WebBook. Over 475 liquid/compressed-water conditions spanning $T = 0$–$350\,°$C and $P = 1$–$2\,000$ bar, the AARE is **1.0%** with a maximum of 5% near the liquid–vapour boundary at 350 °C. For subsurface CO_2 storage conditions ($T \leq 120\,°$C, $P = 75$–600 bar), the AARE reduces to **0.8%**. Full details, the parity plot, and the signed-error surface are provided in Appendix S4.1.

5. Computational Performance

5.1. Flash convergence statistics

5.1.1. Parameter-space scan for the salt-free CPA binary

To rigorously evaluate the robustness and efficiency of the accelerated SSI and multi-guess stability test, we perform a comprehensive parameter-space scan over $86 \times 18 \times 19 = 29{,}412$ conditions spanning $T = 0$–$425\,°$C, $P = 1$–1500 bar, and $z_4 = 0.001$–0.999 (Table 3). At each (T, P, z_4) point, we ran the stability test with all six initial guesses, followed by four independent flash strategies: standard SSI with Wilson K initialization ("Std+Wilson"), accelerated SSI with Wilson K ("Acc+Wilson"), accelerated SSI with K from the stability test ("Acc+StabK"), and the full hierarchical robust algorithm ("Robust").

Table 3: Parameter-space scan grid for the salt-free CPA binary.

Variable	Range	Points	Spacing
T (°C)	0–425	86	5 °C
P (bar)	1–1500	18	semi-log
z_{CO_2}	0.001–0.999	19	variable
Total		29,412	

Of the 29,412 conditions, 19,586 (66.6%) are classified as single-phase by the stability test and 9,825 (33.4%) as two-phase. Table 4 compares six flash strategies on the two-phase subset. The hierarchical robust algorithm achieves 100% convergence by combining the stability-informed K-value initialization with Wilson-K fallback for the 0.3% of points where the stability K fails to converge. Only a single point at the extreme corner of the domain ($T = 425\,°$C, $P = 1500$ bar, $z_{CO_2} = 0.995$) produces a spurious instability; all neighbors are correctly identified as single-phase.

The accelerated SSI reduces the mean iteration count from 25.4 to 13.7 (a 46% reduction, or 1.9× speedup) compared to standard SSI with the same Wilson K initialization (Figs. 6a and S11). The speedup is most pronounced near the CO_2 critical point and at extreme compositions, where standard SSI requires many iterations to converge (Fig. 5). Using K-values from the stability test as the flash initial guess provides a further 8% reduction in iterations (13.7 to 12.6) on average.

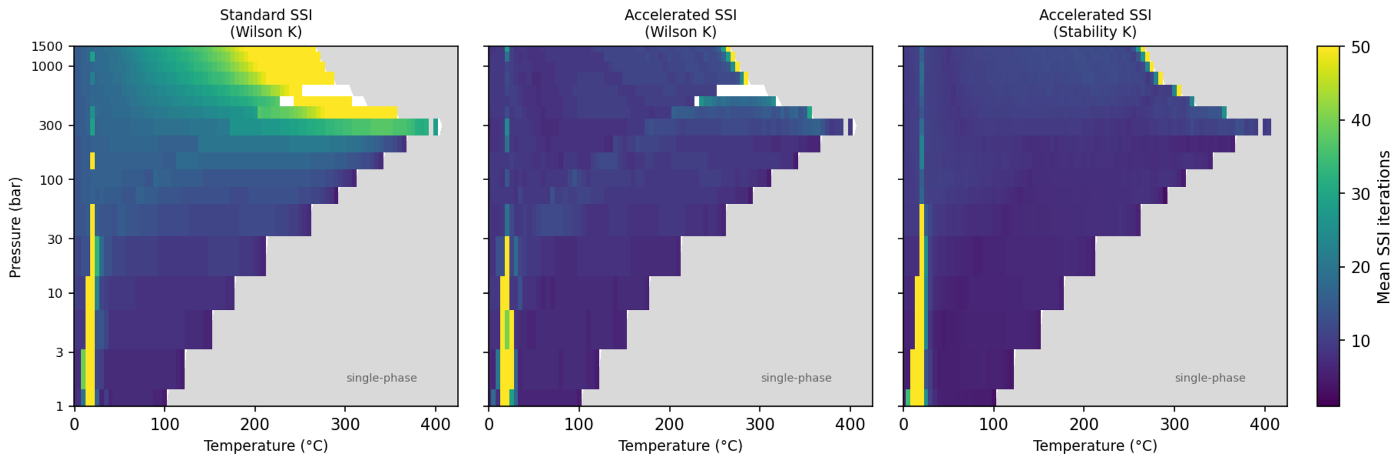


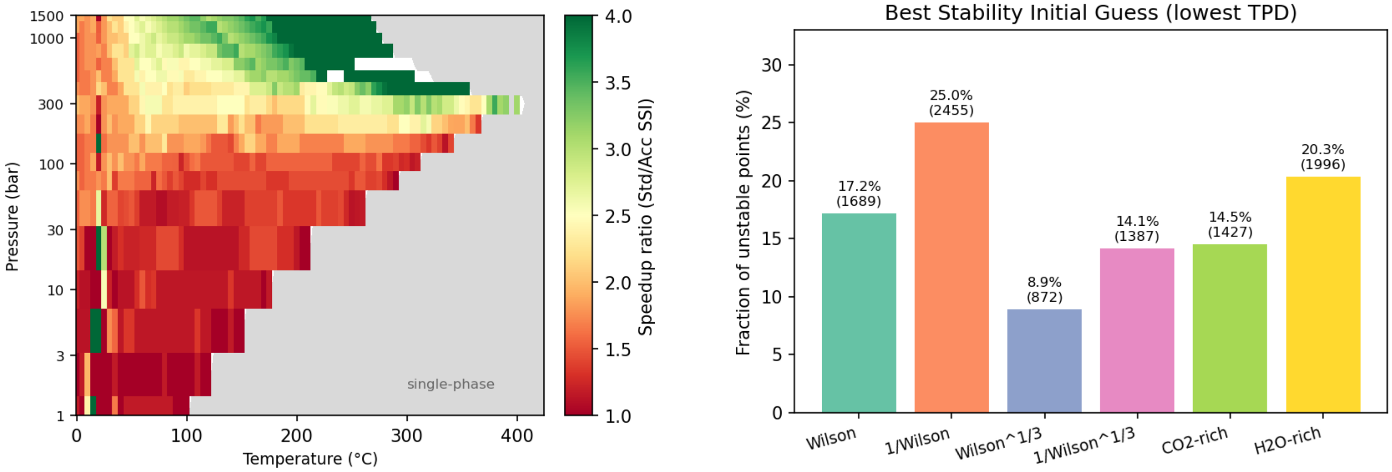
Figure 5: Mean SSI iteration count for the salt-free CPA flash, averaged over all z_{CO_2} values (two-phase points only), as a function of T and P. Left: standard SSI with Wilson K initialization. Center: accelerated SSI with Wilson K. Right: accelerated SSI with stability-test K. The accelerated SSI substantially reduces iteration counts across the entire parameter space, with the largest improvements near the CO_2 critical point ($T \approx 31\,°C$, $P \approx 74\,bar$).

Table 4: Comparison of six flash strategies across 9,825 two-phase conditions from the parameter-space scan (T = 0–425 °C, P = 1–1500 bar, z_4 = 0.001–0.999). The first four rows use cold-start initialization; the fifth and sixth use K-values interpolated from a pre-computed solution table extended to T = 0–400 °C (99.9% of two-phase points). The sixth row additionally applies Newton polish once $\|\mathbf{g}\| < 10^{-4}$, with mean 4.8 SSI and 2.1 Newton iterations per converged call.

Strategy	Converged	Conv. rate	Mean iters	Median iters
Std SSI + Wilson K	9,756	99.3%	25.4	17
Acc SSI + Wilson K	9,775	99.5%	13.7	9
Acc SSI + Stability K	9,793	99.7%	12.6	9
Acc SSI + Stability K + Wilson K fall-back	9,825	**100%**	13.0	9
Acc SSI + Table K	9,818	99.9%	9.8	6
Acc SSI + Table K + Newton	9,808	99.8%	**6.9**	5

When a pre-computed solution table is available—as in sequential simulation where flash solutions from the previous time step or neighboring grid cells serve as initial guesses—the iteration count drops further to a mean of 9.8 (median 6), a 25% improvement over the best cold-start strategy. The table covers T = 0–360 °C with 1 °C resolution and all practically relevant pressures (P = 1–2000 bar); it captures 99.9% of the two-phase scan points, eliminating the Wilson-K fallback for virtually all conditions. Because the table also pre-classifies each cell as single- or two-phase, the Michelsen stability test (six guesses, ≈ 10 ms per guess) can be skipped for all cells confidently inside the two-phase window, so the dominant cost of a cold-start stability+flash call ($\approx 30\times$ higher wall time) is avoided in the warm-start regime. Phase compositions obtained from the two approaches agree to within 1.1% AARE for x_{CO_2} and 3.9% for y_{H_2O}, with residual discrepancies concentrated near the two-phase boundary where both methods are most sensitive to initial guesses.

Adding Newton polish (Table 4, bottom row) reduces the mean count further to 6.9 (4.8 SSI + 2.1 Newton), a 29% improvement over table-warm-started SSI alone and a 73% reduction from the standard cold-start. Newton polish is triggered at 85.6% of points; the remaining 14.4% converge in fewer than $\tau_{\text{switch}} = 10^{-4}$ SSI steps and need no Newton correction. In the near-CO_2-critical region ($T \approx 7$–$57\,^\circ$C) the Newton step can temporarily diverge; the solver then reverts to the pre-Newton SSI state and continues, so no convergence is lost in that region. The iteration breakdown and wall-time ratio across the full (T, P) parameter space are shown in Fig. S10.



(a) Iteration speedup ratio (standard/accelerated SSI)

(b) Distribution of best stability initial guess

Figure 6: (a) Iteration speedup from accelerated SSI, averaged over z_4 (two-phase only). The speedup exceeds $2\times$ across most of the parameter space and reaches 3–$4\times$ near the critical region. (b) Distribution of the initial guess that found the lowest TPD across all 9,825 unstable points. No single guess dominates: each of the six trials is the best for 9–25% of conditions, demonstrating the importance of multiple initial guesses for robust stability testing.

Fig. 6b shows that all six stability initial guesses contribute materially: Wilson is best for 17.2% of unstable points, inverse Wilson for 25.0%, H_2O-rich for 20.3%, CO_2-rich for 14.5%, inverse cube-root for 14.1%, and cube-root Wilson for 8.9%. While the first two guesses (Wilson and inverse Wilson) can detect instability (TPD < 0) for 99.8% of conditions, they locate the *most negative* TPD—and thus the best K-value initialization for the subsequent flash—for only 42.2%. All six guesses are required to ensure the optimal K-value initialization across the full parameter space.

5.1.2. Convergence statistics for the eCPA ternary flash

To characterize eCPA flash performance over the full thermodynamic parameter space relevant to geological CO_2 storage and geothermal energy production, we compute the three-dimensional solution table described in Section 3.6.1, covering the grid summarized in Table 5.

Table 5: Parameter grid for the eCPA ternary solution table (CO_2+H_2O+NaCl, $361 \times 100 \times 14 = 505{,}400$ points).

Dimension	Range	Points
Temperature T	0–360 °C	361 (1 °C steps)
Pressure P	1–2000 bar	100 (log-spaced)
NaCl feed molality m_s	0–6 mol kg^{-1}	14

Each grid point was evaluated with the full hierarchical stability+flash algorithm: the Michelsen stability test (six trial compositions) is performed first; only genuinely unstable conditions proceed to

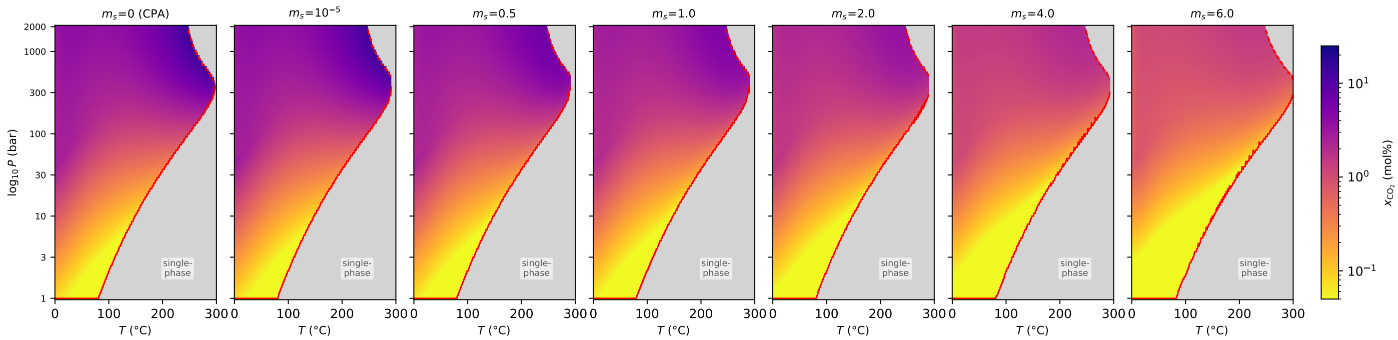


Figure 7: Equilibrium CO_2 mol% in the aqueous phase (x_{CO_2}) over the T–P plane at representative feed $z_4 = 0.5$, for six NaCl molalities (columns, $m_s = 0$–$6\,\mathrm{mol\,kg^{-1}}$). The logarithmic color scale spans three orders of magnitude to resolve both the dilute near-bubble-point regime and the high-solubility hydrothermal limit. Grey regions are thermodynamically single-phase at those conditions. Red curves mark the phase envelope: the solid line is the dew-point boundary (lower pressure limit of the two-phase window) and the dashed line is the bubble-point boundary (upper pressure limit). The phase boundaries are nearly independent of m_s; the salting-out effect manifests as a reduction in x_{CO_2} within the window, not as a shift of its extent.

the K-value SSI flash. The CO_2 feed fraction z_4 is not a table dimension; a representative value $z_4 = 0.5$ is used for the stability call (K-values and tie-line endpoints are only weakly z_4-dependent, as discussed in Section 3.6.1). Of the 505,400 grid points, 36,100 use the CPA sub-model ($m_s = 0$, no electrostatic terms); the eCPA model covers the remaining 469,300 points.

Table 6 reports the outcome of all evaluations. Every grid point is classified—100% convergence is achieved over the complete three-dimensional domain.

Table 6: Classification of eCPA flash outcomes over the full 3D solution table (505,400 total points; 469,300 eCPA, 36,100 CPA).

Classification	Count	Description
Two-phase, converged	289,065	Valid two-phase split; SSI converged
Single-phase (stable)	216,335	Michelsen stability test: no phase split
Total classified	505,400	100%

CO_2 solubility in the aqueous phase. Figures 7 and 8 map x_{CO_2} and y_{H_2O}, respectively, across the full (T, P) domain for six NaCl molalities (and a pure salt-free condition with CPA) at a representative feed $z_4 = 0.5$; red curves trace the dew- (solid) and bubble-point (dashed) boundaries, with the gray region indicating single-phase conditions. Fig. 9 maps mean flash wall time per call across the T–P plane for all six NaCl molalities, serving as a proxy for solver effort. Fig. 7 shows x_{CO_2} (aqueous phase) over a logarithmic color scale spanning 0.05–25 mol%. The two-phase window is broad, spanning the full pressure range from 1 to 2000 bar at most temperatures below ∼250 °C, and closes at the highest temperatures as CO_2 and water approach complete miscibility. Three physically distinct solubility regimes are apparent. At low temperatures ($T \lesssim 50\,°C$), the CO_2-rich phase is a dense supercritical or liquid-like fluid; solubility is moderate and increases with pressure. At intermediate temperatures (50–200 °C), solubility passes through a minimum as the competing effects of pressure and thermal expansion partially cancel. At high temperatures ($T \gtrsim 250\,°C$), solubility rises sharply because the

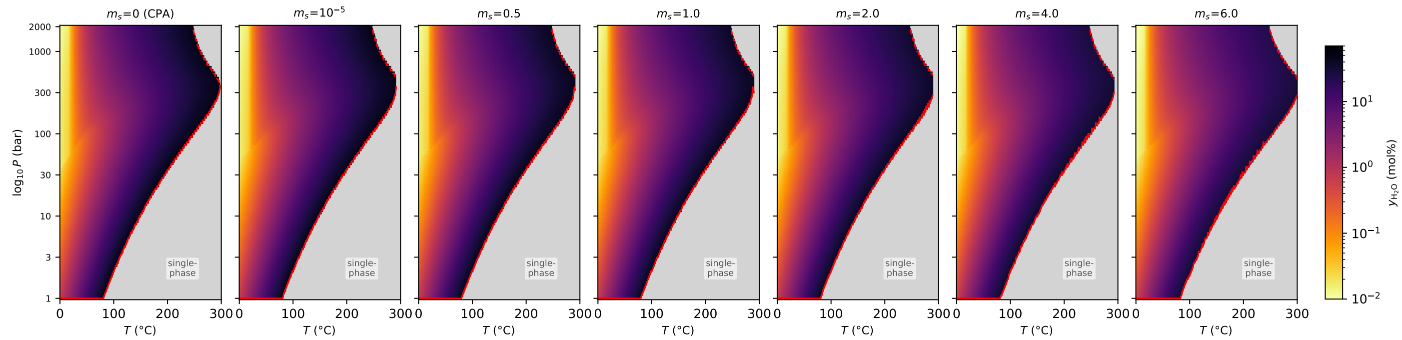


Figure 8: Equilibrium H_2O mol% in the CO_2-rich phase (y_{H_2O}) over the T–P plane at representative feed $z_4 = 0.5$, on the same grid as Fig. 7. Unlike x_{CO_2}, water content in the CO_2-rich phase is governed almost entirely by temperature (water vapour pressure), producing near-vertical isopleths that are nearly independent of pressure and NaCl molality. Red curves show the phase envelope (solid: dew point; dashed: bubble point); their near-identical positions across columns confirm that NaCl has negligible influence on the phase boundaries. At temperatures above 250 °C the CO_2-rich phase contains more than 25 mol% H_2O, approaching 70 mol% at the highest temperatures—a regime relevant to geothermal CO_2 storage.

aqueous phase increasingly resembles steam; values reach 20–25 mol% at the highest pressures, well into a regime relevant to hydrothermal and geothermal CO_2 storage. Increasing NaCl molality (left to right) reduces CO_2 solubility within the two-phase window—the classical salting-out effect. The phase boundaries (red curves) are nearly independent of m_s: the dew-point shifts by less than 15% across the full salinity range, and the bubble-point (visible only near $T \approx 250$ °C where it falls below the 2000 bar grid ceiling) likewise shows negligible salinity dependence. The dominant effect of NaCl is therefore a reduction in x_{CO_2} within the two-phase window rather than a change in the window's extent.

H_2O content of the CO_2-rich phase. Fig. 8 shows y_{H_2O} (CO_2-rich phase) on the same grid layout. In sharp contrast to CO_2 solubility, the water content of the CO_2-rich phase is controlled almost entirely by temperature: the color field in each panel varies primarily along the horizontal axis, with near-vertical isopleths. This reflects the dominant role of the water saturation pressure, which rises steeply with temperature and sets the water activity in the CO_2-rich phase regardless of pressure or salinity. At $T = 50$ °C, y_{H_2O} is below 1 mol% across essentially all pressures; by $T = 300$ °C it reaches 50–70 mol%, meaning the nominal CO_2-rich phase is in fact predominantly water vapour. The effect of NaCl (columns) on y_{H_2O} is weak: the isopleths and the phase envelope (red curves) are essentially unchanged across all salinity columns, confirming that water content in the CO_2-rich phase is set by temperature alone and is insensitive to brine salinity.

Flash wall time. Fig. 9 maps the mean wall time per eCPA flash call (table-warm-started K-value SSI) across the T–P plane for six NaCl molalities; gray regions are single-phase. The $m_s = 0$ (CPA) column is excluded: its cold-start six-trial robust solver runs at a median of ∼215 ms per call (reported in Table 7), which would render uniformly at the colorbar ceiling; the warm-start table applies equally to CPA. Wall time serves as a proxy for solver effort. Times are lowest ($\lesssim 5$ ms) in the interior of the two-phase region, where the warm-started SSI converges in 2–3 steps. Times rise near the phase boundaries—particularly near the bubble point and the CO_2 critical temperature—where K-values approach unity and more iterations are required. The pattern is qualitatively similar across all NaCl molalities.

Table 7 summarizes SSI iteration counts and wall times across the experimental validation datasets used in Sections 4.1–4.2.

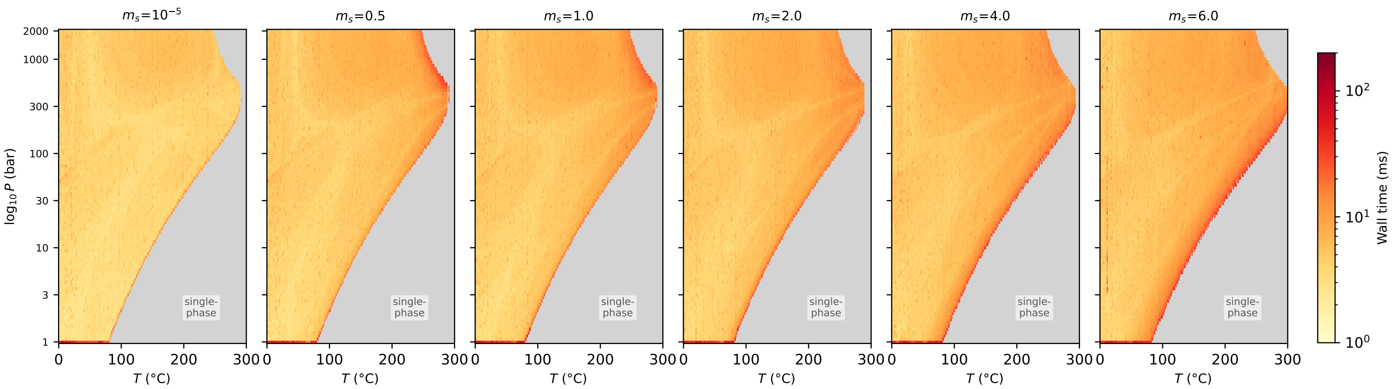


Figure 9: Mean wall time per flash call (ms) for the eCPA warm-started K-value SSI over the T–P plane for six NaCl molalities (columns), from the 3D scan. Gray regions are thermodynamically single-phase. The colormap is log-scaled (1–200 ms). Times are lowest in the interior of the two-phase window and highest near the phase boundaries where K-values approach unity and more SSI iterations are required. The $m_s = 0$ (CPA) column is excluded; its cold-start robust solver runs at a median of $\sim$215 ms per call and is reported separately in Table 7.

Table 7: Flash convergence statistics across the experimental validation datasets.

System	Method	N calls	Mean iter	Median	Max	Median time/call
CO_2+H_2O	CPA (cold, accel.)	29,412	6.3	6	272	$\sim$215 ms
CO_2+H_2O	CPA (warm SSI)	626	6.2	6	40	$\sim$2 ms
CO_2+H_2O	eCPA (warm SSI, $m_s \approx 0$)	631	5.9	6	10	$\sim$2 ms
CO_2+NaCl	eCPA (warm SSI)	686	5.8	6	16	$\sim$2 ms

All three warm-start rows use the same solution table and achieve a median of 6 SSI iterations and $\lesssim$2 ms per call. The warm-started CPA row directly demonstrates that the $\sim$100× speedup over cold-start CPA (row 1) is attributable entirely to the warm-start strategy: the eCPA model adds electrostatic terms not present in CPA, yet both achieve comparable wall times when started from a good initial guess. Among the eCPA rows, the additional salt does not materially increase cost. Each warm-started SSI step costs only $\sim$0.2 ms; the solution-table lookup itself takes $\lesssim$0.05 ms and contributes negligible overhead. Convergence is robust across the full validation range.

5.2. Inner EoS solve efficiency: Newton–Raphson versus cold-start alternatives

Each outer SSI iteration requires solving the inner per-phase EoS system. The two systems — CPA's (Z, χ_w, χ_c) and eCPA's $(Z_w, \varepsilon_r, \chi_w)$ — differ in structure but share the same warm-started Newton–Raphson strategy described in Sections 3.3 and 3.4; the analytical Jacobians are derived in Appendices S3.1 and S3.2. The benchmarks below quantify the advantage of the warm-started Newton solve over the respective cold-start alternatives, and are distinct from the outer SSI+Newton-polish acceleration discussed in Section 5.1 and Fig. S10.

CPA: (Z, χ_w, χ_c) system. Starting from the previous SSI iterate (typically within 0.5% of the converged value), the warm-started Newton solve converges in **3–4 iterations** (median 3, maximum 4 across 9,825 two-phase conditions). The cold-start fallback — a 12-point log-spaced scan to bracket the root of F_1, followed by Brent refinement — requires on average 23 residual evaluations (median 21) per call, equivalent to $\approx$0.38 ms. The warm-start Newton solve costs $\approx$0.03 ms across 128 two-phase conditions spanning $T = 27$–$227\,°$C and $P = 50$–800 bar: a **13**$\times$ reduction in inner-solve cost per SSI step.

eCPA: $(Z_w, \varepsilon_r, \chi_w)$ system. For the aqueous 3×3 system, a 0.5–1% warm-start perturbation yields convergence in **4–6 iterations** (median 5 at 1%), with a wall time of $\approx$0.25 ms per call. The cold-start alternative — solving from a heuristic initial guess via a general-purpose modified-Powell root-finder — requires on average 47 residual evaluations and costs $\approx$1.5 ms per call: a **6**$\times$ overhead.

Computational cost of CPA and eCPA versus a cubic EOS. To quantify the per-iteration overhead of the association terms relative to a plain cubic EoS, we timed each component of a single SSI iteration across 871 two-phase (T, P) conditions sampled from the CO_2–H_2O parameter space. CPA (with the Newton Z/χ solve) was compared against a notional SRK reference (same EoS parameter computation and fugacity evaluation, but Z found by direct solution of the cubic and $\chi \equiv 1$). The per-iteration median wall times were 98 µs (CPA) versus 32 µs (SRK), a ratio of **3.1**$\times$. The EoS-parameter and fugacity-evaluation steps are identical between the two models; the entire overhead arises in the Z/χ solve, which costs 75 µs (CPA Newton, two phases) versus 9 µs (SRK direct cubic, two phases). This factor-of-3 per-iteration cost must be weighed against the additional physics captured by CPA: accurate partitioning of a self- and cross-associating system over a wide temperature and pressure range, and (in the full eCPA model) correct salt activity without empirical brine corrections.

5.3. Simplified versus full flash: accuracy and efficiency

Over 3,285 two-phase conditions spanning $T = 10$–$250\,°$C, $P = 25$–800 bar, and $m_s = 1$–$6\,\mathrm{mol\,kg^{-1}}$, the simplified flash reproduces the full K-value SSI result with AARE $< 2\%$ in dissolved CO_2 molality for $T \leq 80\,°$C, rising to 3.2% at $100\,°$C and exceeding 5% above $120\,°$C as the $y_{H_2O} = 0$ assumption deteriorates. Phase identification agrees with the full flash in 99.7% of cases. The accuracy is insensitive to brine salinity: the error is controlled entirely by temperature. Contrary to expectation, the mean wall time is essentially identical to the full flash (0.8$\times$ speedup), because Brent root-finding on the 1D fugacity residual requires $\sim$11 inner Newton evaluations while SSI needs only $\sim$6 outer iterations with at most two inner solves each. Full per-temperature statistics, diagnostic figures, and a salinity breakdown are provided in Appendix S4.2.

6. Reservoir Simulation Demonstration

To confirm that the flash algorithms developed in this work perform reliably under realistic conditions, we implement them within an isothermal IMPEC compositional simulator for CO_2 injection into heterogeneous 3D saline aquifers. Across 300 time steps and all 2500 grid cells of a 50×50 domain, no flash failures are recorded in either the salt-free (CPA, $m_s = 0$) or saline (eCPA, $m_s = 4\,\mathrm{mol\,kg^{-1}}$) case, confirming 100% convergence of the hierarchical stability and flash approach. The warm-started eCPA flash achieves a throughput of 14,270 calls $\mathrm{s^{-1}}$ and the CPA flash 8,840 calls $\mathrm{s^{-1}}$, with the flash accounting for 80–87% of total step cost. Full details of the numerical scheme, simulation setup, spatial results, and performance metrics are provided in Appendix S6.

7. Conclusions

This work presents the first complete phase stability and flash framework built on the eCPA equation of state of Coelho et al. (2025), extending it from single-phase property evaluation to fully converged, production-ready VLE calculations for CO_2 + H_2O + NaCl mixtures across reservoir to hydrothermal conditions.

The central algorithmic contribution is the tight integration of four independently novel elements: (i) fast inner Newton–Raphson solvers for the CPA and eCPA root-finding problems, based on analytical Jacobians derived in closed form, which reduce the per-evaluation overhead to only $\sim 3\times$ that of a simple cubic EoS; (ii) a hierarchical outer flash algorithm combining Michelsen stability testing with six initial guesses, accelerated SSI, and optional Newton polish — achieving 100% convergence over a comprehensive parameter-space scan; (iii) a precomputed three-dimensional (T, P, m_s) solution table that provides warm starts reducing flash iterations by a factor of three relative to cold-start SSI; and (iv) a simplified flash that exploits the $y_{\mathrm{H_2O}} \approx 0$ approximation to reduce the equilibrium problem to a single scalar equation, yielding errors below 2% in dissolved CO_2 for $T \leq 80\,^\circ\mathrm{C}$ at essentially the same computational cost. Together, these enable warm-started eCPA flash at throughputs competitive with conventional cubic-EoS implementations, as demonstrated in a prototype IMPEC reservoir simulator.

Validation against >1100 experimental data points shows AARE of 6.5% for CO_2 solubility under subsurface conditions, 6.6% for CO_2 molality in NaCl brine, and 0.76% for aqueous-phase density without any volume-shift correction — the last being a direct consequence of the CPA a_0/b parameterization against pure-water liquid density. Extended comparison against the IAPWS-95 reference equation over 475 liquid conditions ($T = 0$–$350\,^\circ\mathrm{C}$, $P = 1$–$2000\,\mathrm{bar}$) confirms an overall density AARE of 1.0%, falling to 0.8% for subsurface storage conditions.

Future Work

The algorithms and validation presented here were implemented entirely in Python. While this facilitates rapid prototyping and transparent benchmarking, it means that the absolute throughput numbers reported are conservative relative to what a compiled implementation would achieve. The natural next step is to integrate the CPA and eCPA flash routines into a production-level compiled simulator that fully accounts for gravity-driven and density-driven flow, Fickian diffusion, capillarity, and explicitly resolved discrete fractures in three-dimensional unstructured grids (e.g. Moortgat et al. 2011b,a; Moortgat and Firoozabadi 2013, 2016; Moortgat 2018; Solovský and Firoozabadi 2025, 2026). Such an integration will enable high-fidelity simulation of CO_2 sequestration in saline aquifers at reservoir scale.

A second, more fundamental extension is the generalization to brines containing multiple salts (e.g. NaCl + $CaCl_2$ + $MgCl_2$, as encountered in deep formation waters). In the single-salt eCPA framework the NaCl molality provides a single scalar coupling between phase amounts and phase compositions. For multi-salt brines the ionic interactions become more complex, and it will likely be necessary to shift from a molecular-species basis to an ionic basis, treating each cation and anion as an independent component with its own association and electrostatic parameters. Extending the eCPA framework in this direction represents a significant but tractable step toward a truly general thermodynamic engine for saline CO_2 storage.

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Data and Code Availability

All source code implementing the CPA and eCPA equations of state, the phase stability and flash algorithms, the solution table construction, and the reservoir simulator described in this work is freely available at `https://github.com/XXX/ecpa-flash` (*upon acceptance of paper). The repository includes the precomputed three-dimensional solution table, all experimental VLE data used for validation, and scripts to reproduce every figure and table in this paper.

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S1. Nomenclature

Symbols are listed in alphabetical order within each group. Phase subscripts and superscripts are defined at the end of the table.

Latin symbols

Symbol	Description	Units
Upper case		
A	Dimensionless SRK attractive parameter, $A = a(T)P/(RT)^2$	—
A^r	Residual Helmholtz energy per mole	J mol^{-1}
A^{assoc}	Wertheim association term of A^r	J mol^{-1}
A^{Born}	Born solvation contribution to A^r	J mol^{-1}

continued on next page

Symbol	Description	Units
A^{DH}	Debye–Hückel electrostatic contribution to A^r	J mol^{-1}
A^{SRK}	Soave–Redlich–Kwong contribution to A^r	J mol^{-1}
AARE	Average absolute relative error	%
B	Dimensionless SRK co-volume parameter, $B = bP/(RT)$	—
D_c	CO_2 association denominator	—
D_w	H_2O association denominator	—
F_1, F_2, F_3	CPA inner-solve residuals in (Z, χ_w, χ_c)	—
F_z	Molar fractional flow of CO_2	—
G^{res}	Residual Gibbs energy	J mol^{-1}
G_0, G_1, G_2	eCPA inner-solve residuals in $(Z_w, \varepsilon_r, \chi_w)$	—
H_{ij}	Jacobian entries for the eCPA $(Z_w, \varepsilon_r, \chi_w)$ system	various
J_{ij}	Jacobian entries for the CPA (Z, χ_w, χ_c) system	various
K	Isotropic permeability	mD
K_i	Vapour–liquid K-value of species i, $K_i = y_i/x_i$	—
K_i^W	Wilson K-value correlation	—
L	Step-size clipping limit in accelerated SSI	—
L_x, L_y, L_z	Reservoir domain dimensions	m
M_w	Molar mass of water ($0.018015\,\mathrm{kg\,mol^{-1}}$)	kg mol^{-1}
N_A	Avogadro number	mol^{-1}
N_x, N_y	Number of grid cells in x- and y-direction	—
P	Pressure	bar
P_c	Critical pressure	bar
P_{BHP}	Bottom-hole pressure	bar
P_{sat}	Saturation pressure of pure H_2O	bar
R	Universal gas constant	J mol^{-1} K^{-1}
R_B	Born radius of ion	Å
S_g	CO_2-rich phase saturation	—
S_{cw}	CO_2–H_2O cross-association strength ratio	—
S_{gr}	Residual gas saturation (Corey)	—
S_{wc}	Connate water saturation (Corey)	—
T	Temperature	°C or K
T_c	Critical temperature	K
$\mathrm{TPD}(\mathbf{W})$	Tangent-plane distance function	—
V_g	Radial distribution function derivative with respect to η	—
W	Second derivative of V_g; also unnormalized trial composition vector	—
Z	Compressibility factor, $Z = Pv/(RT)$	—
Z_c	Compressibility factor of the CO_2-rich phase	—
Z_w	Compressibility factor of the aqueous phase	—
Lower case		
$a(T)$	Mixture SRK attractive parameter	J cm^3 mol^{-2}
a_{0i}	Pure-component reference attractive parameter	J cm^3 mol^{-2}

continued on next page

Symbol	Description	Units
b	Mixture SRK co-volume parameter	$cm^3\ mol^{-1}$
b_i	Pure-component co-volume	$cm^3\ mol^{-1}$
c_i	Péneloux volume-shift parameter of species i	$cm^3\ mol^{-1}$
c_{1i}	Soave temperature-dependence coefficient of species i	—
d_i	Reference chemical-potential offset in TPD, $\ln z_i + \ln \hat{\phi}_i(\mathbf{z})$	—
$g(\eta)$	Carnahan–Starling radial distribution function	—
g_w	Kirkwood g-factor for water	—
$\mathbf{g}^{(k)}$	Fugacity-ratio residual vector at SSI iteration k	—
k_{ij}	Binary interaction parameter between species i and j	—
k_B	Boltzmann constant	$J\ K^{-1}$
k_{rg}, k_{rw}	Corey relative permeabilities (CO_2-rich, aqueous)	—
$m^{(k)}$	Dominant-eigenvalue acceleration factor at iteration k	—
m_c	CO_2 molality in the aqueous phase	$mol\ kg^{-1}$
m_{feed}	NaCl feed molality	$mol\ kg^{-1}$
m_s	NaCl molality (feed; solution-table grid coordinate)	$mol\ kg^{-1}$
m_s^{aq}	Self-consistent equilibrium NaCl molality in the aqueous phase	$mol\ kg^{-1}$
n_g, n_w	Corey exponents (CO_2-rich, aqueous)	—
q	Well volumetric flow rate	$m^3\ s^{-1}$
r_w	Wellbore radius	m
t	Time	yr
v	Molar volume	$cm^3\ mol^{-1}$
w_i	Normalised trial mole fractions in stability test	—
x_i	Mole fraction of species i in the aqueous phase	—
$x^{\text{sf}}_{CO_2}$	CO_2 mole fraction on a salt-free basis	—
y_i	Mole fraction of species i in the CO_2-rich phase	—
$\mathbf{z}$	Overall feed composition vector, $\mathbf{z} = (z_1, z_4)$	—
z_1	Overall H_2O feed mole fraction, $z_1 = 1 - z_4$	—
z_4	Overall CO_2 feed mole fraction	—

Greek symbols

Symbol	Description	Units
α_{is}	NRTL non-randomness parameter (ion–solvent)	—
β	CO_2-rich phase molar fraction	—
$\delta^{A_iB_j}$	Association bonding volume between sites A_i and B_j	$m^3\ mol^{-1}$
ΔU_{is}	Ion–solvent interaction energy (eCPA NRTL term)	$J\ mol^{-1}$
ε/k_B	Association well depth (temperature units)	K
ε_0	Vacuum permittivity	$F\ m^{-1}$
ε_∞	High-frequency (optical) relative permittivity	—
ε_r	Static relative permittivity (dielectric constant) of water	—
η	Packing fraction, $\eta = b/(4v)$	—
$\hat{\phi}_i$	Fugacity coefficient of species i	—
κ	Association volume parameter in the Wertheim term	—
μ_{0w}	Permanent dipole moment of water	C m
μ_g, μ_w	Dynamic viscosity (CO_2-rich, aqueous)	Pa s
ρ	Molar density	$mol\ m^{-3}$
σ	Collision diameter (ion hard-sphere radius)	Å
$\tau, \tau_{\text{switch}}$	SSI convergence tolerance; Newton-polish switch threshold	—
ϕ	Porosity	—
χ^{A_i}	Fraction of molecules of species i not bonded at site A	—
χ_w, χ_c	Non-bonded fractions of H_2O and CO_2 (CPA)	—
χ_{1w}, χ_{1c}	Non-bonded fractions at site 1 (eCPA)	—
ω_i	Acentric factor of species i	—

Subscripts and superscripts

Symbol	Meaning
c	CO_2-rich phase; or CO_2 species
w	Aqueous phase; or H_2O species
i, j	Species index
1, 4	Species index in K-value flash (CO_2 = 1, H_2O = 4)
aq	Aqueous phase
assoc	Association contribution
Born	Born solvation contribution
DH	Debye–Hückel electrostatic contribution
feed	Feed (overall) quantity
phys	Physical (SRK) contribution
res	Residual thermodynamic quantity
SRK	Soave–Redlich–Kwong reference fluid
c, i	Critical property of species i

Note on species indices. In the K-value flash formulation (Section 3.4), species are indexed numerically: CO_2 = 1, H_2O = 4, Na^+ = 2, Cl^- = 3. In the CPA flash formulation (Section 3.3), species are denoted by letters: c (CO_2), w (H_2O).

S2. EoS Parameters

For completeness, Tables B.9 and B.10 reproduce the key EoS parameters used in this work, all from Coelho et al. (2025).

Table B.9: Pure-component CPA/eCPA parameters (Coelho et al., 2025). For ions, $a_0 = 0$. H_2O uses the 4C association scheme.

Species	a_0 ($\mathrm{J\,m^3\,mol^{-2}}$)	c_1	b ($\mathrm{cm^3\,mol^{-1}}$)	ε/k_B (K)	κ	σ (Å)	R_B (Å)
H_2O	0.1228	0.6736	14.52	2003	κ^*	—	—
CO_2	0.3508	0.7602	27.20	—	—	—	—
Na^+	0	—	16.49	—	—	2.356	1.665
Cl^-	0	—	40.83	—	—	3.187	1.828

$^*\kappa = \kappa_{\text{assoc}} \cdot b_{H_2O}$, $\kappa_{\text{assoc}} = 69.2 \times 10^{-3}$.

Table B.10: Temperature-dependent binary interaction parameters (Coelho et al., 2025). $X = A_X(T/T_{c,c})^2 + B_X(T/T_{c,c}) + C_X$, $T_{c,c} = 31.3\,°\mathrm{C}$.

Parameter	A	B	C
k_{cw} (CO_2–H_2O binary interaction parameter)	−0.492	2.101	−1.571
S_{cw} (cross-assoc. strength)	+0.192	−0.173	−0.009

Ion–neutral pair	ΔU^{ref} ($\mathrm{J\,mol^{-1}}$)	α (K)	T^α (K)	Source
H_2O–NaCl (ΔU_{ws})	−1858	1573	340	Maribo-Mogensen et al. (2015)
CO_2–NaCl (ΔU_{cs})	+6056	692	244	Coelho et al. (2025)

S3. Analytical Jacobians for the Inner Newton Solvers

Within each SSI iteration two distinct Newton systems are solved, one per phase. The CO_2-rich phase and both phases of the salt-free CPA binary use a 3×3 system in (Z, χ_w, χ_c) (Section S3.1). The eCPA aqueous phase requires a different 3×3 system in $(Z_w, \varepsilon_r, \chi_w)$, because the relative permittivity ε_r satisfies an additional Kirkwood self-consistency equation (Section S3.2).

S3.1. CPA binary: Newton system for (Z, χ_w, χ_c)

The three fixed-point residuals are

$$F_1 = Z - \frac{Z}{Z - B} + \frac{A}{Z + B} - G\big[x_w(\chi_w - 1) + x_c(\chi_c - 1)\big] = 0, \tag{C.1}$$

$$F_2 = \chi_w - \frac{Z}{D_w} = 0, \tag{C.2}$$

$$F_3 = \chi_c - \frac{Z}{D_c} = 0, \tag{C.3}$$

with the denominators

$$D_w = Z + 2x_w\chi_w\Delta_w + 2x_c\chi_c\Delta_{cw}, \qquad D_c = Z + 2x_w\chi_w\Delta_{cw}, \tag{C.4}$$

where $\Delta_w = g(\eta)\kappa_w[\exp(\varepsilon_w/k_BT) - 1]P/(RT)$, $\Delta_{cw} = S_{cw}\Delta_w$, $\eta = B/(2Z)$, and $G = 1 + V_g$ with

$$V_g \equiv \frac{\eta g'(\eta)}{g(\eta)} = \frac{\eta(5-2\eta)}{(1-\eta)(2-\eta)}, \qquad W \equiv \frac{\mathrm{d}V_g}{\mathrm{d}\eta} = \frac{10 - 8\eta + \eta^2}{(1-\eta)^2(2-\eta)^2}, \tag{C.5}$$

for the Carnahan–Starling radial distribution function $g(\eta) = (2-\eta)/[2(1-\eta)^3]$. Because $\eta \propto Z^{-1}$, the key identity is

$$Z\,\frac{\partial\Delta_w}{\partial Z} = -V_g\,\Delta_w, \tag{C.6}$$

from which all Z-derivatives of D_w and D_c follow. Setting $D = x_w(\chi_w - 1) + x_c(\chi_c - 1)$, the 3×3 Jacobian $J_{ij} = \partial F_i/\partial v_j$ with $\mathbf{v} = (Z, \chi_w, \chi_c)^\top$ is:

Row 1.

$$J_{11} = 1 + \frac{B}{(Z-B)^2} - \frac{A}{(Z+B)^2} + \frac{D\,W\eta}{Z}, \tag{C.7}$$

$$J_{12} = -G\,x_w, \tag{C.8}$$

$$J_{13} = -G\,x_c. \tag{C.9}$$

Row 2. Using $Z\partial D_w/\partial Z = Z - 2(x_w\chi_w + x_c\chi_cS_{cw})V_g\Delta_w$ and $\chi_w = Z/D_w$:

$$J_{21} = -\frac{2\Delta_w(1+V_g)(x_w\chi_w + x_c\chi_cS_{cw})}{D_w^2}, \tag{C.10}$$

$$J_{22} = 1 + \frac{2x_w\Delta_w\chi_w}{D_w}, \tag{C.11}$$

$$J_{23} = \frac{2x_cS_{cw}\Delta_w\chi_w}{D_w}. \tag{C.12}$$

Row 3. Using $Z\partial D_c/\partial Z = Z - 2x_w\chi_wS_{cw}V_g\Delta_w$ and $\chi_c = Z/D_c$:

$$J_{31} = -\frac{2x_w\chi_wS_{cw}\Delta_w(1+V_g)}{D_c^2}, \tag{C.13}$$

$$J_{32} = \frac{2x_wS_{cw}\Delta_w\chi_c}{D_c}, \tag{C.14}$$

$$J_{33} = 1. \tag{C.15}$$

Note $J_{33} = 1$ because D_c does not depend on χ_c; Newton therefore has a direct update for χ_c at each step. The same Jacobian structure applies to the CO_2-rich phase with the CO_2-rich-phase compositions and parameters.

S3.2. eCPA aqueous phase: Newton system for $(Z_w, \varepsilon_r, \chi_w)$

In the aqueous phase the relative permittivity ε_r is an additional unknown satisfying the Kirkwood self-consistency equation (C.17). The CO_2 cross-association fraction χ_c is eliminated algebraically via $\chi_c = Z_w/D_c$ with $D_c = Z_w + 2x_w\chi_w S_{cw}\Delta_w$, reducing the system to three equations in $\mathbf{v} = (Z_w, \varepsilon_r, \chi_w)^\top$:

$$G_0 = Z_w - Z_w^{\text{phys}} - Z_w^{\text{assoc}} - Z_w^{\text{DH}} - Z_w^{\text{Perm}} = 0, \tag{C.16}$$

$$G_1 = T_1(\varepsilon_r, \varepsilon_\infty) - T_2(Z_w, \chi_w) = 0, \tag{C.17}$$

$$G_2 = \chi_w - \frac{Z_w}{D_w} = 0, \tag{C.18}$$

where T_1 and T_2 are the left- and right-hand sides of the Onsager–Kirkwood relative permittivity fixed-point equation (Coelho et al., 2025), $D_w = Z_w + 2x_w\chi_w\Delta_w + 2x_c\chi_c S_{cw}\Delta_w$, and all Z_w-derivatives of Δ_w satisfy the same identity (C.6) (with Z_w in place of Z).

Row 2 (exact). G_2 has the same structure as F_2 (C.2) with ε_r absent from D_w, so

$$H_{20} = -\frac{2\Delta_w(1+V_g)(x_w\chi_w + x_c\chi_c S_{cw})}{D_w^2}, \tag{C.19}$$

$$H_{21} = 0, \tag{C.20}$$

$$H_{22} = 1 + \frac{2x_w\Delta_w\chi_w}{D_w}. \tag{C.21}$$

Row 1 (exact). $T_1 = F(\varepsilon_r, \varepsilon_\infty)$ (left-hand side of the Onsager–Kirkwood equation) and

$$T_2 = (N_A c/9\varepsilon_0 k_B T)x_w g_w \mu_{0w}^2$$

depend on $(Z_w, \varepsilon_r, \chi_w)$ through $\varepsilon_\infty(Z_w)$, ε_r, and the Kirkwood g-factor $g_w(\chi_w)$, respectively:

$$H_{10} = \left(\frac{\partial F}{\partial \varepsilon_\infty}\right)\frac{V(\partial\varepsilon_\infty/\partial V)}{Z_w} - \frac{N_A\mu_{0w}^2}{9\varepsilon_0 k_B T}\, x_w \left(-\frac{\rho}{Z_w}\, g_w + \rho\,\frac{\partial g_w}{\partial Z_w}\right), \tag{C.22}$$

$$H_{11} = \frac{\partial F}{\partial \varepsilon_r} = \frac{2\varepsilon_r^2 + \varepsilon_\infty^2}{\varepsilon_r^2(\varepsilon_\infty + 2)^2}, \tag{C.23}$$

$$H_{12} = -\frac{N_A\rho\mu_{0w}^2}{9\varepsilon_0 k_B T}\, x_w\, \frac{\partial g_w}{\partial \chi_w}, \tag{C.24}$$

where $\rho = P/(Z_w RT)$, $V(\partial\varepsilon_\infty/\partial V)$ follows from the Clausius–Mossotti relation for ε_∞ (Coelho et al., 2025), and $\partial g_w/\partial Z_w$ and $\partial g_w/\partial \chi_w$ follow from differentiating the Kirkwood g-factor (Coelho et al., 2025) through $\mathcal{P}_{ww}$ and $\mathcal{P}_{wc}$ using Eq. (C.6) for $\partial\delta_w/\partial Z_w$ and $\partial\chi_c/\partial\chi_w = -Z_w \cdot 2x_w S_{cw}\Delta_w/D_c^2$, respectively.

Row 0 (partially approximate). The Debye screening length satisfies

$$\frac{\partial\kappa}{\partial Z_w} = -\frac{\kappa}{2Z_w}, \qquad \frac{\partial\kappa}{\partial\varepsilon_r} = -\frac{\kappa}{2\varepsilon_r}. \tag{C.25}$$

The ε_r-column entry combines DH and permittivity contributions,

$$H_{01} = -\frac{\partial Z_w^{\text{DH}}}{\partial\varepsilon_r} - \frac{\partial Z_w^{\text{Perm}}}{\partial\varepsilon_r}, \tag{C.26}$$

where $\partial Z_w^{\mathrm{DH}}/\partial\varepsilon_r$ is obtained from the DH compressibility expression (Coelho et al., 2025) through $\kappa(\varepsilon_r)$ using Eq. (C.25), and $\partial Z_w^{\mathrm{Perm}}/\partial\varepsilon_r$ requires differentiating $Z_w^{\mathrm{Perm}} = -a_{\varepsilon_r} V(\partial\varepsilon_r/\partial V)$ with respect to ε_r through κ and through $V(\partial\varepsilon_r/\partial V)$, using the second-order F-derivatives

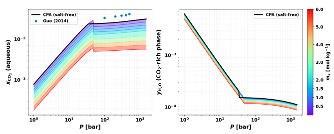
$$\frac{\partial^2 F}{\partial\varepsilon_r^2} = -\frac{2\varepsilon_\infty^2}{\varepsilon_r^3(\varepsilon_\infty+2)^2}, \qquad \frac{\partial^2 F}{\partial\varepsilon_r\,\partial\varepsilon_\infty} = \frac{4(\varepsilon_\infty-\varepsilon_r^2)}{\varepsilon_r^2(\varepsilon_\infty+2)^3}. \tag{C.27}$$

The χ_w-column entry is

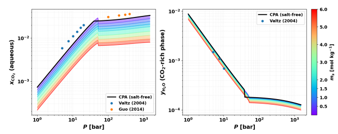
$$H_{02} = -\frac{\partial Z_w^{\mathrm{assoc}}}{\partial\chi_w} - \frac{\partial Z_w^{\mathrm{Perm}}}{\partial\chi_w}, \tag{C.28}$$

where $\partial Z_w^{\mathrm{assoc}}/\partial\chi_w$ follows from the Wertheim association compressibility (Coelho et al., 2025) through $\chi_c(\chi_w)$, and $\partial Z_w^{\mathrm{Perm}}/\partial\chi_w$ is obtained by differentiating the chi cross-derivative system with respect to χ_w and propagating through $\mathcal{P}_{ww}$, $\mathcal{P}_{wc}$, g_w, and $V(\partial F/\partial V)$. The Z_w-column entry H_{00} approximates $\partial(Z_w^{\mathrm{phys}} + Z_w^{\mathrm{assoc}} + Z_w^{\mathrm{DH}} + Z_w^{\mathrm{Perm}})/\partial Z_w$ by omitting the implicit Z_w-dependence of $V(\partial\chi_w/\partial V)$ and $V(\partial\chi_c/\partial V)$ through the chi cross-derivative system. Comparison with centered finite differences confirms this approximation incurs a 1–2 % error in H_{00} at typical conditions, which does not impair Newton convergence from warm starts (the other eight entries are exact to machine precision).

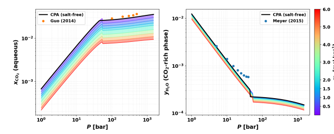
S4. Additional Validation Figures



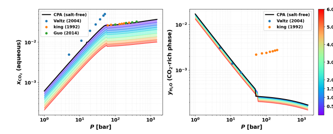
(a) 0 °C (273 K)



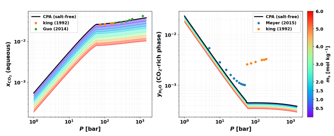
(b) 5 °C (278 K)



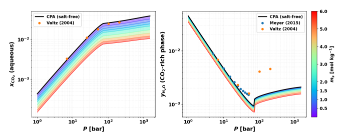
(c) 10 °C (283 K)



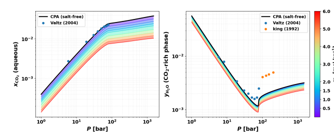
(d) 15 °C (288 K)



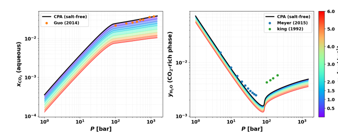
(e) 20 °C (293 K)



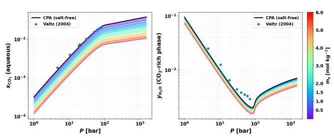
(f) 31 °C (304 K)



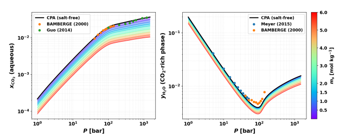
(g) 35 °C (308 K)



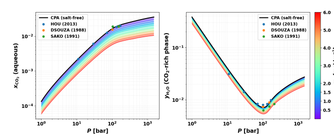
(h) 40 °C (313 K)



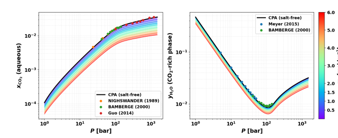
(i) 45 °C (318 K)



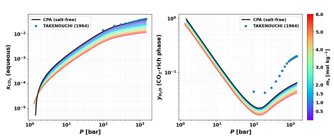
(j) 60 °C (333 K)



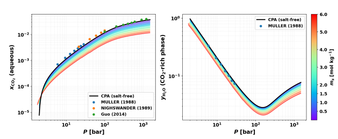
(k) 75 °C (348 K)



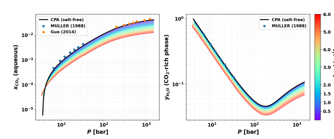
(l) 80 °C (353 K)



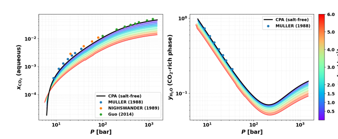
(m) 110 °C (383 K)



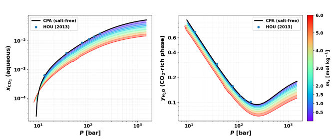
(n) 120 °C (393 K)



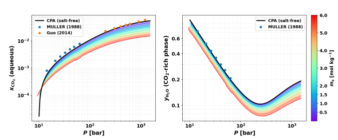
(o) 140 °C (413 K)



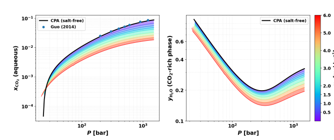
(p) 160 °C (433 K)



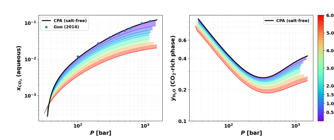
(q) 175 °C (448 K)



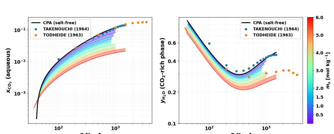
(r) 180 °C (453 K)



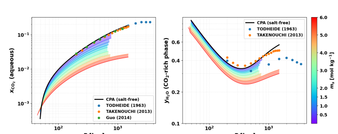
(s) 220 °C (493 K)



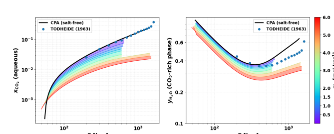
(t) 240 °C (513 K)



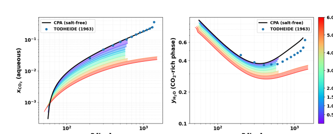
(u) 260 °C (533 K)



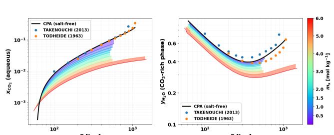
(v) 265 °C (538 K)



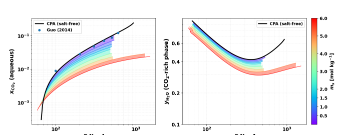
(w) 268 °C (541 K)



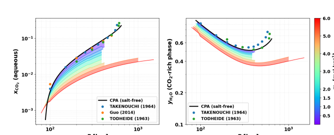
(x) 270 °C (543 K)



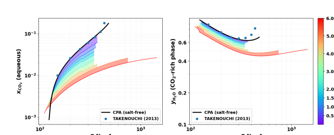
(y) 275 °C (548 K)



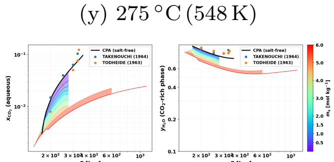
(z) 280 °C (553 K)



(aa) 300 °C (573 K)

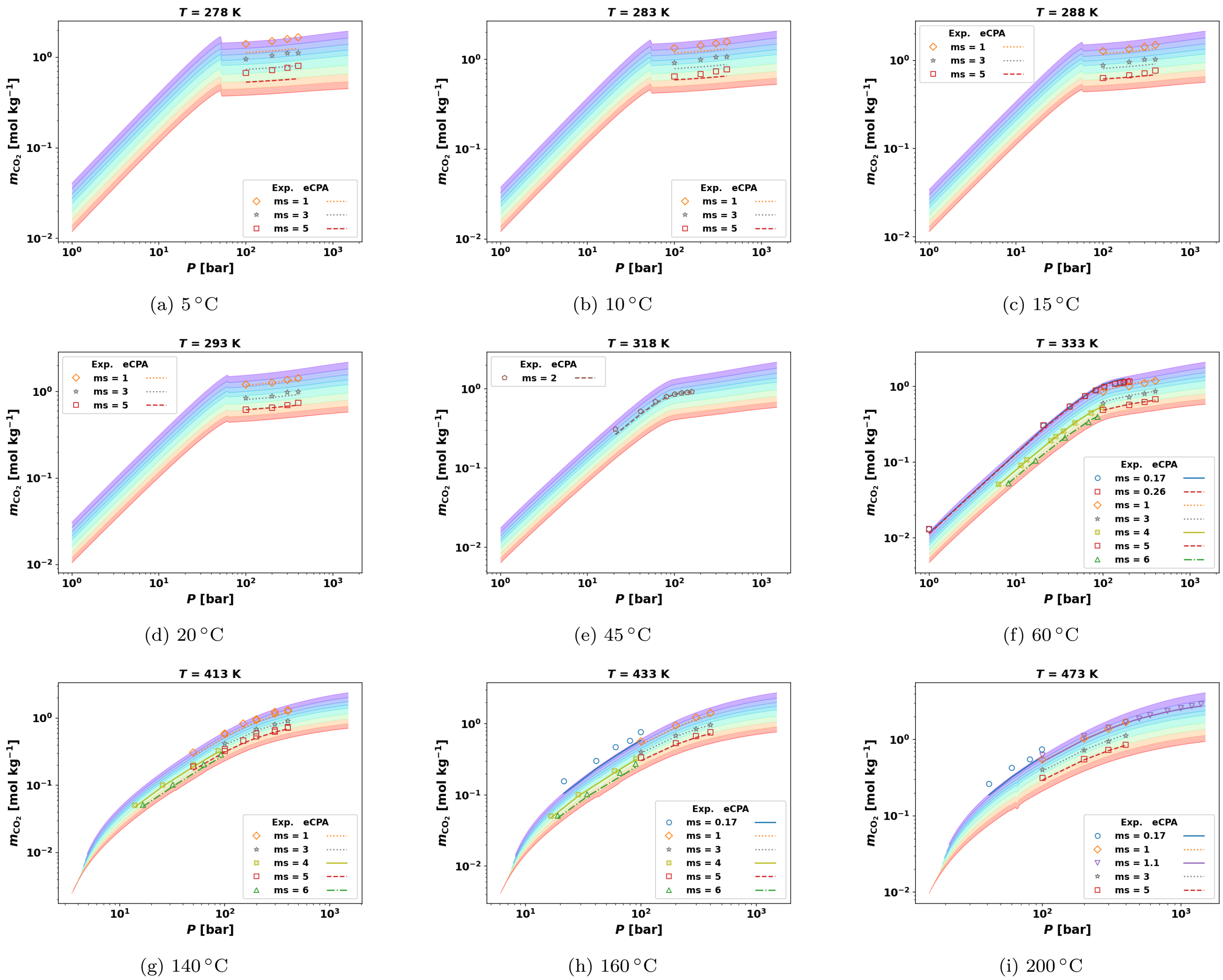


(ab) 325 °C (598 K)



(ac) 350 °C (623 K)

Figure S1: CO_2 + H_2O phase equilibria at all 29 temperatures not shown in Fig. 1 (0–350 °C), 4 panels per row. Each panel shows the CO_2 mole fraction in the aqueous phase (x_{CO_2}, left sub-panel) and H_2O mole fraction in the CO_2-rich phase (y_{H_2O}, right sub-panel) vs. pressure. Open symbols: experimental data (Coelho et al., 2025); black solid line: CPA (salt-free); rainbow-colored filled bands: eCPA spanning $m_s = 0$–$6\,\mathrm{mol\,kg^{-1}}$ (violet to red), with discrete molality contours overlaid.

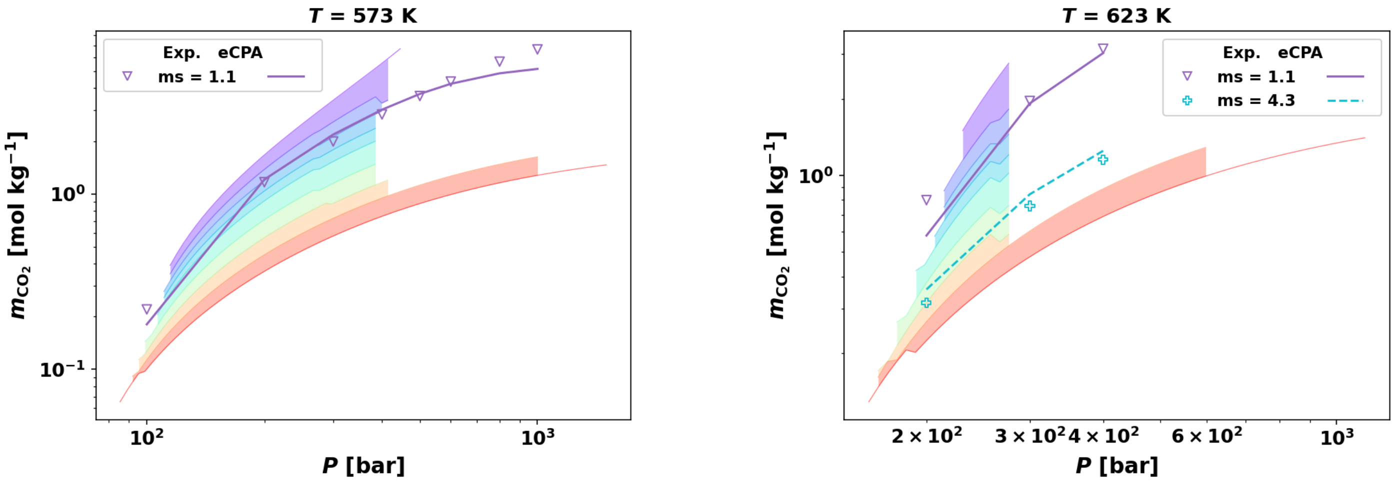


(a) 5 °C (b) 10 °C (c) 15 °C

(d) 20 °C (e) 45 °C (f) 60 °C

(g) 140 °C (h) 160 °C (i) 200 °C

Figure S2: CO_2 solubility in NaCl brine at nine representative temperatures not shown in Fig. 2 (5–200 °C), including data at high NaCl molalities up to $m_s = 6\,\mathrm{mol\,kg^{-1}}$ from multiple sources. Each panel shows CO_2 molality m_c [$\mathrm{mol\,kg^{-1}}$] vs. pressure (log–log). Open symbols: experimental data (Guo et al., 2015; Rumpf et al., 1994; Bando et al., 2003; Chabab et al., 2020; Liu et al., 2021, 2011; Nighswander et al., 1989; Takenouchi and Kennedy, 1965; Yan et al., 2011); discrete lines with markers: eCPA at the experimental NaCl molalities. Rainbow-colored background bands: continuous eCPA predictions across $m_s = 0$–$6\,\mathrm{mol\,kg^{-1}}$ (color scale as in Fig. 1).



(a) 300 °C

(b) 350 °C

Figure S3: CO_2 solubility in NaCl brine at hydrothermal temperatures (a) $T = 300\,°C$, (b) $T = 350\,°C$. Each panel shows CO_2 molality m_c [mol kg^{-1}] vs. pressure. Open symbols: experimental data (Takenouchi and Kennedy, 1965); lines with markers: eCPA predictions at the experimental NaCl molalities. At $T \geq 400\,°C$, eCPA predicts single-phase (fully miscible) behavior for all available experimental pressures (300–900 bar).

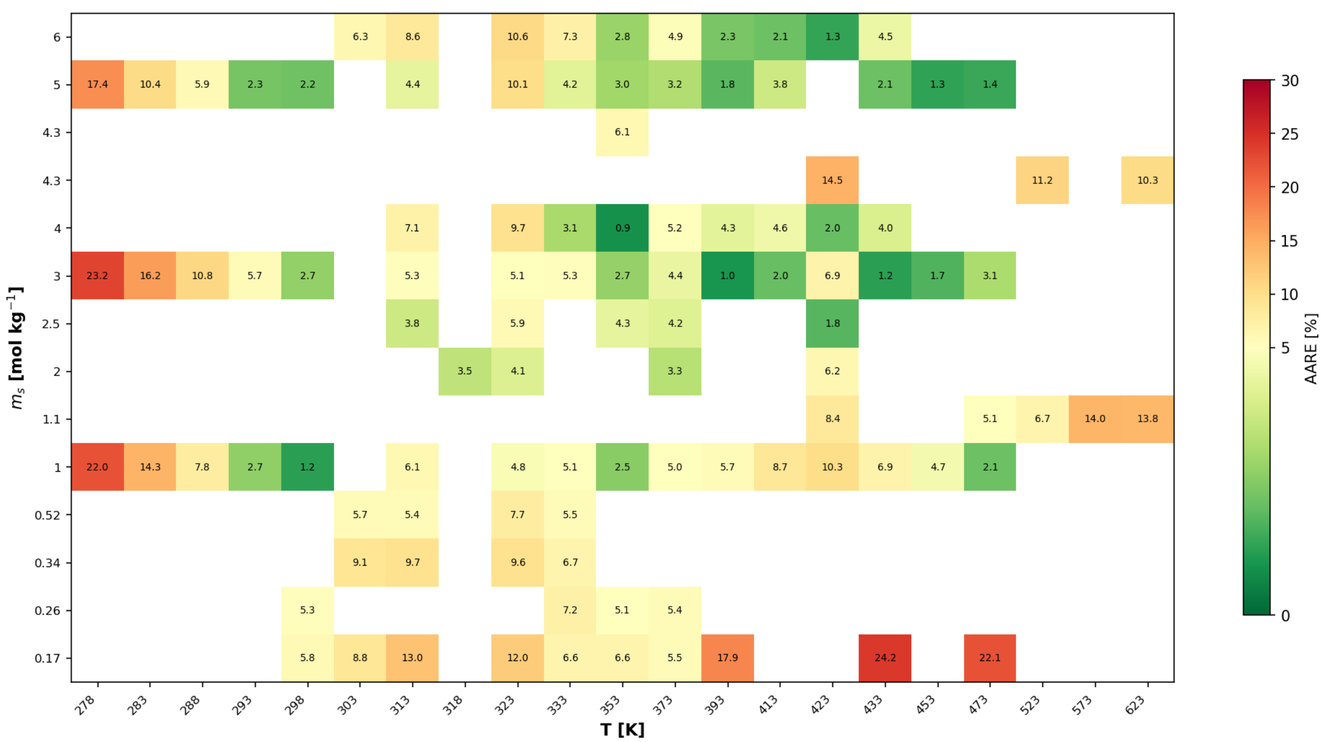


Figure S4: Heatmap of absolute relative error (AARE, %) for the CO_2 + NaCl validation dataset ($T = 5$–$350\,°C$), showing all individual NaCl molalities present in the experimental data. Color scale is diverging at 5%: green $\leq 5\%$, red $\geq 10\%$. Empty cells indicate no experimental data at that (T, m_s) combination.

S4.1. Aqueous-phase density: extended validation

Comparison with CO_2-saturation experiments. Figure S5 shows the eCPA parity plot against 37 CO_2-saturated aqueous-density measurements from King et al. (1992) and Nighswander et al. (1989) ($T = 15$–$200\,°C$), reproducing the 0.76% AARE of Coelho et al. (2025) without any Péneloux volume shift for H_2O. CPA results at $m_s = 0$ are indistinguishable from eCPA and are omitted for clarity.

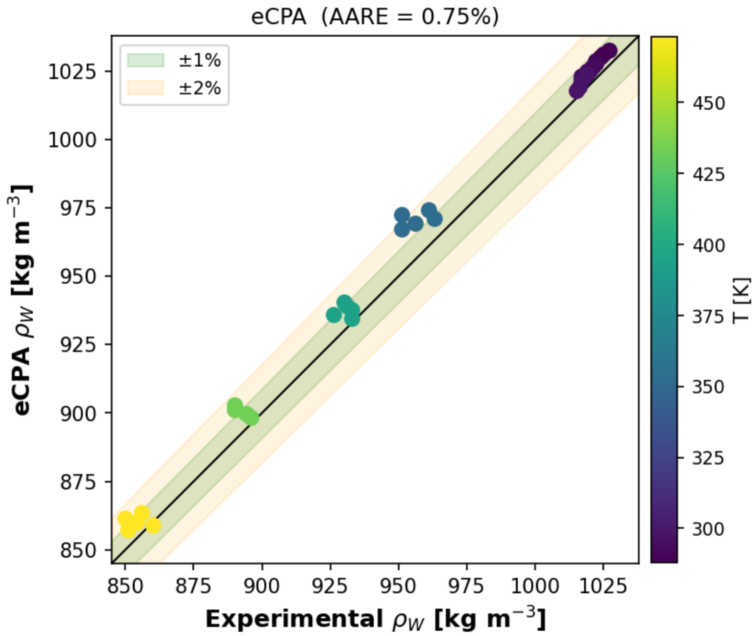


Figure S5: Aqueous-phase density of CO_2-saturated water ($m_s = 0$) predicted by eCPA versus experimental data from King et al. (1992) and Nighswander et al. (1989) (37 points, $T = 15$–$200\,°C$). Points are colored by temperature; shaded bands show ±1% and ±2% deviations. No Péneloux volume shift is applied to H_2O; the 0.76% AARE reflects the liquid-density accuracy intrinsic to the CPA parameterization.

Comparison with IAPWS-95 over the full paper range. Because CO_2-saturation density measurements cover only a limited (T, P) window, we extend the density validation by comparing pure-water ($m_s = 0$) eCPA predictions against the IAPWS-95 reference equation (Wagner and Pruß, 2002) over a systematic grid of 26 temperatures ($T = 0$–$350\,°C$) and 20 pressures ($P = 1$–$2\,000$ bar), restricted to conditions where IAPWS-95 classifies water as liquid or compressed liquid (475 points total; vapour conditions are excluded as the aqueous-phase EoS is not intended for steam). The eCPA density is computed from the converged aqueous-phase compressibility Z_w as $\rho = M_{H_2O} \,/\, (Z_w RT/P)$, with no Péneloux correction.

The signed error surface (Fig. S6) reveals a smooth, small-amplitude pattern: eCPA slightly over-predicts density near $0\,°C$ (+0.7%), crosses zero near $15\,°C$, under-predicts by up to −1.6% in the 50–$150\,°C$ range, and recovers toward zero at high T and high P. The one outlying feature is a positive spike reaching 5% near the liquid–vapour boundary at $T \approx 350\,°C$ and moderate pressures, where liquid water becomes highly compressible and the SRK backbone is most challenged. Overall statistics: AARE = 1.0%, bias = −0.7%, maximum ARE = 5.0% (475 liquid conditions). For subsurface CO_2 storage conditions ($T \leq 120\,°C$, $P = 75$–600 bar, $N = 136$), the AARE reduces to 0.8%.

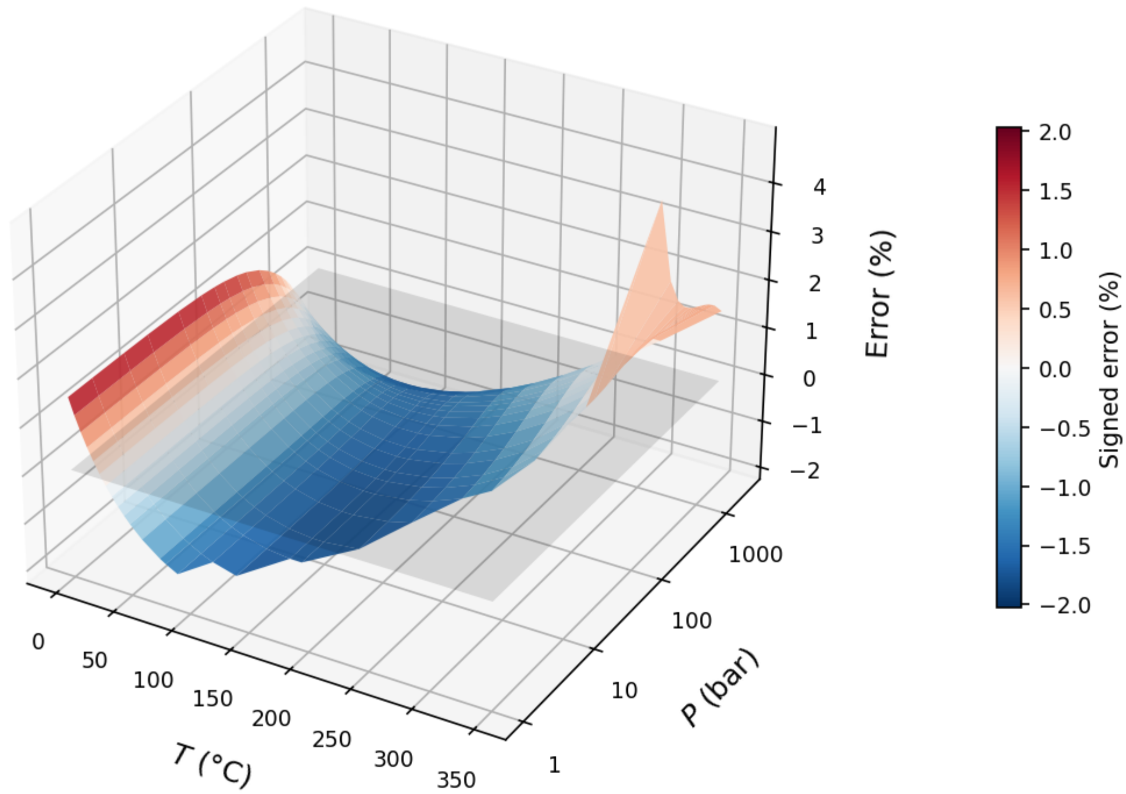


Figure S6: Signed relative error of eCPA pure-water density predictions versus the IAPWS-95 reference equation (Wagner and Pruß, 2002) over $T = 0$–$350\,^\circ$C and $P = 1$–$2\,000$ bar (475 liquid/compressed-water conditions; vapour excluded). The P-axis is logarithmic. The transparent grey plane marks zero error. Blue regions indicate conditions where eCPA slightly over-predicts density (near $0\,^\circ$C); red regions where it slightly under-predicts (50–$150\,^\circ$C); the narrow red spike near $350\,^\circ$C at moderate pressures corresponds to the highly compressible near-saturation liquid where cubic-EoS predictions are least accurate.

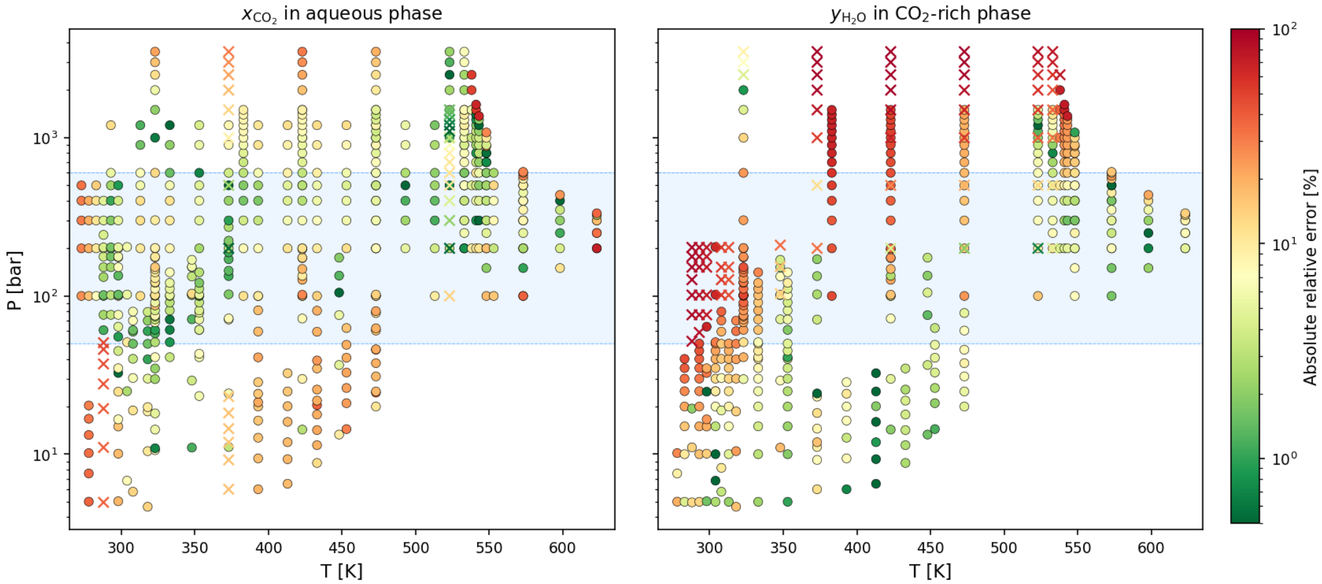


Figure S7: Absolute relative error of eCPA predictions for the CO_2 + H_2O binary as a function of temperature and pressure. Left: CO_2 solubility in the aqueous phase (x_{CO_2}). Right: H_2O content in the CO_2-rich phase (y_{H_2O}). Marker color indicates error magnitude (green < 5%, yellow ≈ 10%, red > 30%); crosses mark points flagged as outliers by cross-reference consistency analysis. The blue shaded band highlights the pressure range most relevant to subsurface CO_2 storage (P = 50–600 bar).

S4.2. Simplified flash: detailed accuracy and efficiency

Table D.11 reports the AARE in dissolved CO_2 molality m_c and the mean y_{H_2O} in the CO_2-rich phase from the full flash as a function of temperature, evaluated over 3,285 two-phase conditions spanning $P = 25$–800 bar, $z_{CO_2} = 0.01$–0.70, and $m_s = 1$–$6\,\mathrm{mol\,kg^{-1}}$. The two quantities track each other closely: the simplified flash introduces an error in m_c that equals, to a good approximation, the water content it neglects.

Table D.11: Accuracy of the simplified flash ($y_{H_2O} = 0$) relative to the full K-value SSI flash, evaluated over 3,285 two-phase conditions spanning $P = 25$–$800\,\mathrm{bar}$, $z_{CO_2} = 0.01$–0.70, and $m_s = 1$–$6\,\mathrm{mol\,kg^{-1}}$. AARE: average absolute relative error in CO_2 molality m_c; $\langle y_{H_2O} \rangle$: mean H_2O mole fraction in the CO_2-rich phase from the full flash (proxy for the assumption error).

T (K)	T (°C)	N	AARE in m_c (%)	$\langle y_{H_2O} \rangle$ (mol%)
283	10	239	0.03	0.02
298	25	239	0.11	0.08
313	40	240	0.37	0.27
323	50	240	0.62	0.45
333	60	240	0.93	0.68
348	75	239	1.55	1.15
353	80	240	1.80	1.34
373	100	238	3.16	2.35
393	120	239	5.37	3.91
413	140	237	8.95	6.19
423	150	237	11.67	7.72
453	180	235	25.49	14.01
473	200	231	45.85	19.79
523	250	191	121.28	33.51

Figure S8 summarises the AARE in m_c, the mean y_{H_2O} from the full flash, and the wall-time speedup as functions of temperature. Figure S9 breaks the AARE down by NaCl molality; the error curves for all salinities are nearly coincident, confirming that the accuracy of the $y_{H_2O} = 0$ approximation is controlled entirely by temperature.

The mean wall time is essentially identical to the full flash (speedup $0.8\times$): Brent root-finding requires on average 11.3 inner Newton evaluations, whereas the full SSI needs 6.4 outer iterations each involving at most two inner solves. The inner Newton cost per call is similar in both paths, so the Brent overhead offsets the gain from eliminating the outer salt iteration. The principal benefits of the simplified flash are therefore conceptual — the equilibrium reduces to a single scalar CO_2-solubility equation — and robustness in salt-free settings where the ternary outer salt loop is absent by construction.

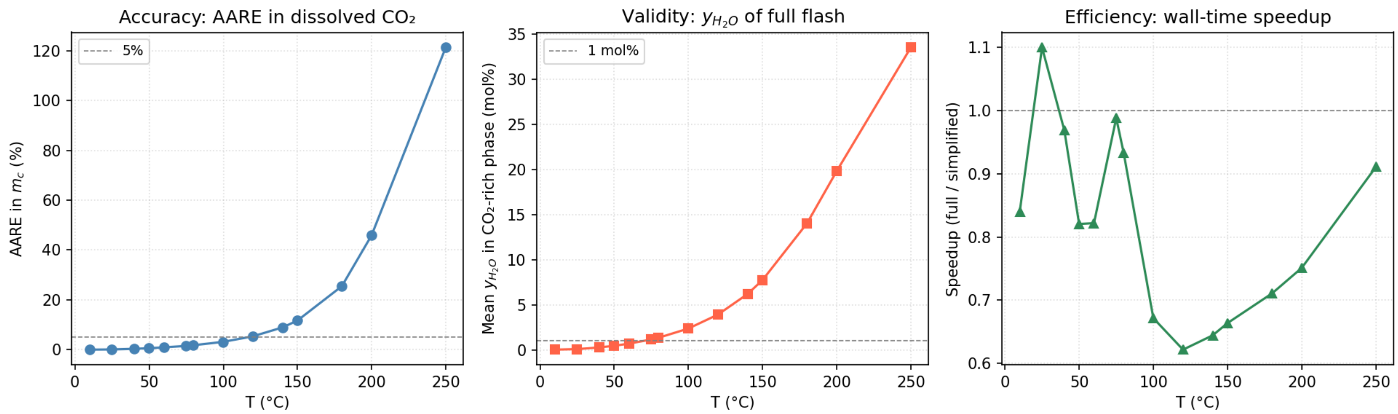


Figure S8: Simplified flash benchmark over 3,285 two-phase conditions (P = 25–800 bar; z_{CO_2} = 0.01–0.70; m_s = 1–6 mol kg^{-1}). **Left**: AARE in dissolved CO_2 molality m_c as a function of temperature; the dashed line marks 5%. **Middle**: mean H_2O mole fraction in the CO_2-rich phase from the full flash, which directly quantifies the magnitude of the $y_{H_2O} = 0$ assumption; dashed line at 1 mol%. **Right**: ratio of mean full-flash wall time to simplified-flash wall time; the two methods are essentially iso-cost across all temperatures.

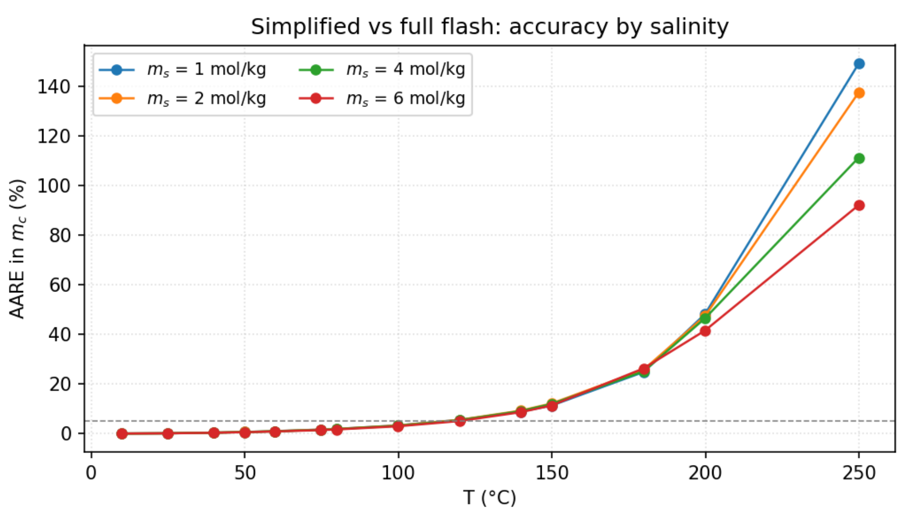


Figure S9: AARE in dissolved CO_2 molality m_c of the simplified flash versus the full K-value SSI, broken down by NaCl molality (m_s = 1–6 mol kg^{-1}). The error curves for different salinities are nearly coincident, confirming that the accuracy of the $y_{H_2O} = 0$ approximation is controlled entirely by temperature and is insensitive to brine salinity. The dashed line marks 5%.

S5. Computational Performance Figures

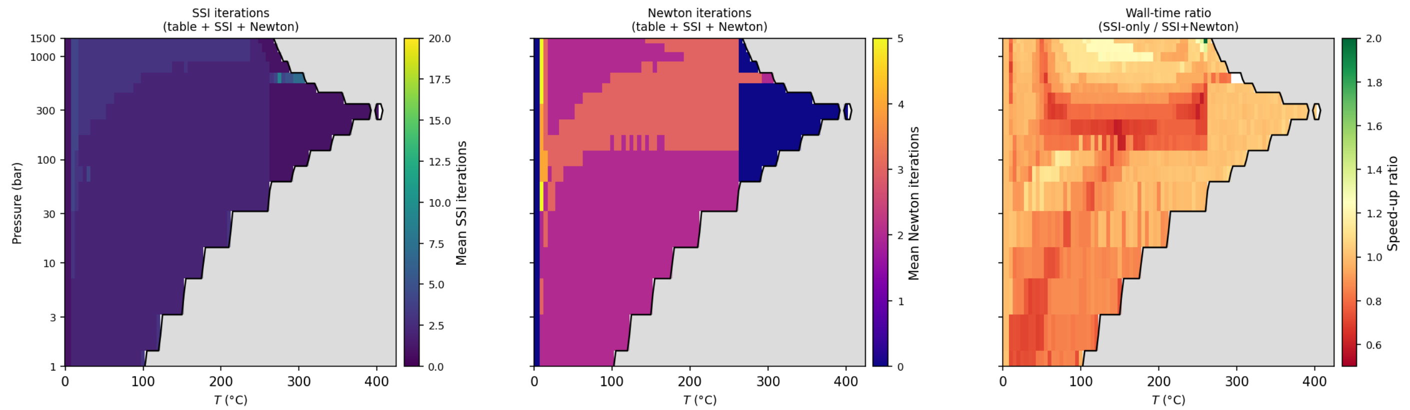


Figure S10: SSI + Newton polish iteration breakdown and wall-time ratio across the parameter-space scan (9,825 two-phase points, averaged over z_{CO_2}). *Left*: mean SSI iteration count before Newton polish is triggered ($\|\mathbf{g}\| < 10^{-4}$); the near-CO_2-critical region ($T \approx 7$–$57\,°C$) requires the most SSI iterations even with table warm-start. *Center*: mean Newton iteration count; most of the domain converges in 2–3 Newton steps after 1–5 SSI steps, while the critical region (blue) often reverts to SSI before Newton is beneficial. *Right*: wall-time ratio (SSI-only / SSI+Newton); values > 1 (green) indicate that Newton polish is faster; values < 1 (red) indicate that the SSI-only path is faster (primarily near the CO_2 critical point where Newton reversion adds overhead).

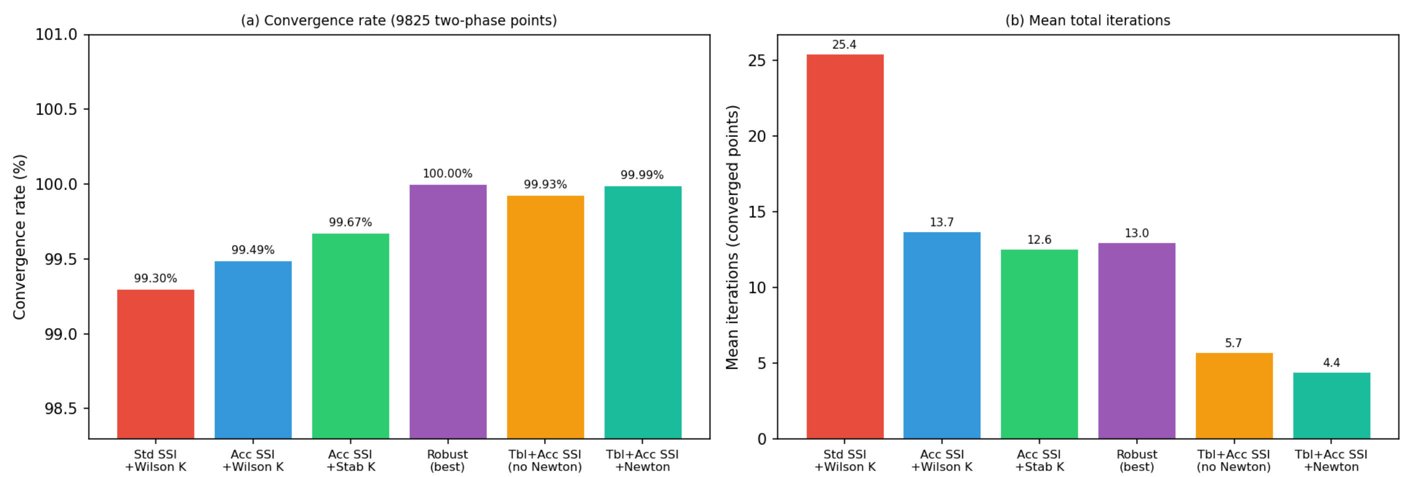


Figure S11: CPA flash strategy comparison across all 9,825 two-phase conditions of the salt-free $CO_2 + H_2O$ parameter-space scan ($T = 0$–$425\,°C$, $P = 1$–$1500\,bar$, $z_4 = 0.001$–0.999). (a) Convergence rate; (b) mean total iterations (converged points only). Strategies are ordered from simplest cold-start (standard SSI with Wilson K) to the most optimized (table warm-start with SSI and Newton polish). The final strategy achieves a 73% reduction in mean iterations relative to the standard cold-start baseline.

S6. Reservoir Simulation Details

Numerical scheme

Pressure is solved implicitly at each time step on a uniform Cartesian grid using either a two-point flux approximation (TPFA, Aavatsmark 2002) or a lowest-order mixed finite element (MFE, RT_0)

scheme (Chavent and Jaffre, 1986). Both were implemented and tested; results shown here use MFE, which avoids the pronounced grid-orientation artefacts of TPFA on Cartesian grids while remaining computationally inexpensive (Moortgat et al., 2011b; Moortgat and Firoozabadi, 2013, 2016). The solved pressure field yields Darcy velocities that drive explicit first-order upwind advection of the overall CO_2 mole fraction z_4. The transport sub-step size is constrained by a CFL condition based on the maximum slope of the molar fractional flow $\partial F/\partial z_4$, estimated by a one-dimensional scan at startup. Production wells are controlled by a fixed bottom-hole pressure (BHP); the injector delivers a fixed volumetric rate.

At each time step, the full Michelsen stability test followed by the two-phase flash is called at every grid cell, using the cell pressure from the previous time step (standard IMPES lagged-pressure). For the eCPA case ($m_s > 0$) the warm-started K-value flash with hierarchical stability fallback is used (Section 3.6); for the salt-free case ($m_s = 0$) the CPA warm-started flash is used. NaCl is non-volatile; the equilibrium aqueous molality m_s^{aq} returned by the flash reflects local salting-out and is used directly in the saturation calculation.

We deliberately replace the true compressible volume-rate divergence by the incompressible constraint $\nabla \cdot \mathbf{u} = q$. This neglects buoyancy, compressibility, and the density difference between injected and displaced fluid. We state this limitation explicitly because neglecting it removes the main physical driver of CO_2 plume geometry in real aquifer storage, and readers should not interpret the results as quantitative predictions. The purpose is solely to confirm that the stability and flash routines converge reliably and efficiently at every cell of a 50×50 grid over 300 time steps spanning 5 years.

Simulation setup

Two simulations are run on identical grids with identical flow parameters, differing only in brine salinity:

- **Case A (CPA):** salt-free brine, $m_s = 0\,\mathrm{mol\,kg^{-1}}$
- **Case B (eCPA):** saline brine, $m_s = 4\,\mathrm{mol\,kg^{-1}}$ NaCl

The domain is a $100\times100\,\mathrm{m}^2$ square of 10 m thickness discretized onto a 50×50 Cartesian grid ($N = 2500$ cells) with homogeneous permeability of 100 mD and porosity $\phi = 0.20$. A five-spot well pattern is used: a CO_2 injector at the domain center and four BHP-controlled producers at the corners (BHP = 150 bar). CO_2 is injected at 20% of total pore volume per year. The simulation runs for 5 years ($T = 77\,^\circ\mathrm{C}$) with 300 pressure time steps. Full simulation parameters are listed in Table F.12.

Table F.12: Reservoir simulation parameters used in Section 6.

Parameter	Symbol	Value
Grid and geometry		
Domain size (x, y)	$L_x \times L_y$	$100 \times 100\,\mathrm{m}$
Domain thickness	L_z	$10\,\mathrm{m}$
Grid cells (x, y)	$N_x \times N_y$	50×50
Rock properties		
Permeability (isotropic, uniform)	K	$100\,\mathrm{mD}$
Porosity	ϕ	0.20
Relative permeability (Corey)		
Connate water saturation	S_{wc}	0.15
Residual gas saturation	S_{gr}	0.05
End-point k_{rg}	k_{rg}^0	0.80
End-point k_{rw}	k_{rw}^0	1.00
Corey exponent (CO_2-rich phase)	n_g	2.0
Corey exponent (aqueous phase)	n_w	2.5
Fluid properties		
Temperature (isothermal)	T	$77\,^\circ\mathrm{C}$ ($350\,\mathrm{K}$)
CO_2-rich phase viscosity	μ_g	$5 \times 10^{-5}\,\mathrm{Pa\,s}$
Aqueous phase viscosity	μ_w	$4.5 \times 10^{-4}\,\mathrm{Pa\,s}$
Well and injection parameters		
Well pattern	—	five-spot
Injector location	—	domain center
Producer locations	—	four corners
Wellbore radius	r_w	$0.10\,\mathrm{m}$
Producer BHP	P_{BHP}	$150\,\mathrm{bar}$
Injection rate	—	20% $\mathrm{PVI\,yr^{-1}}$
Simulation control		
Total simulation time	$t_{\max}$	$5\,\mathrm{yr}$
Number of pressure time steps	N_t	300
Pressure solver	—	MFE (RT_0)
Transport scheme	—	first-order upwind, CFL-limited
Cases		
Case A (CPA)	m_s	$0\,\mathrm{mol\,kg^{-1}}$ (salt-free)
Case B (eCPA)	m_s	$4\,\mathrm{mol\,kg^{-1}}$ NaCl

Results

Fig. S12 shows the final-time ($t = 5\,\mathrm{yr}$) spatial fields for both cases: the CO_2-rich saturation S_g (top row), the dissolved CO_2 molality m_c in the aqueous phase (middle row), and the pressure field (bottom row). The CO_2 saturation maps are strikingly similar between the two cases. This is physically expected: at a high injection rate of 20% $\mathrm{PVI\,yr^{-1}}$, advection strongly dominates over dissolution-driven transport. The CO_2 displacement front is governed primarily by the mobility ratio and the imposed pressure gradient, not by whether the equilibrium CO_2 solubility in the aqueous phase is slightly reduced by salt. Given that CO_2 solubility in water is already low—on the order of 1–2 $\mathrm{mol\,kg^{-1}}$ under typical storage conditions—the further reduction from salting-out has negligible impact on the high-rate displacement dynamics on these time scales.

The dissolved CO_2 maps (middle row) tell a different story. Inside the plume the aqueous phase is everywhere CO_2-saturated at the local equilibrium value: $m_c \approx 1.25\,\mathrm{mol\,kg^{-1}}$ for the salt-free case and only $m_c \approx 0.63\,\mathrm{mol\,kg^{-1}}$ at $m_s = 4\,\mathrm{mol\,kg^{-1}}$—a reduction of approximately a factor of two, consistent with the salting-out effect seen in the validation data (Section 4.2). The white region around the central injector marks cells where the brine has been fully expelled into the CO_2-rich phase (complete drying out, $S_g \to 1$), despite a 15% connate/residual water saturation. Correctly capturing this drying-out regime requires a robust stability and flash model that fully captures the mutual solubilities of CO_2 and H_2O.

No flash failures were recorded in either simulation across all 300 time steps and all 2500 grid cells, confirming the reliability of the hierarchical stability and flash approach.

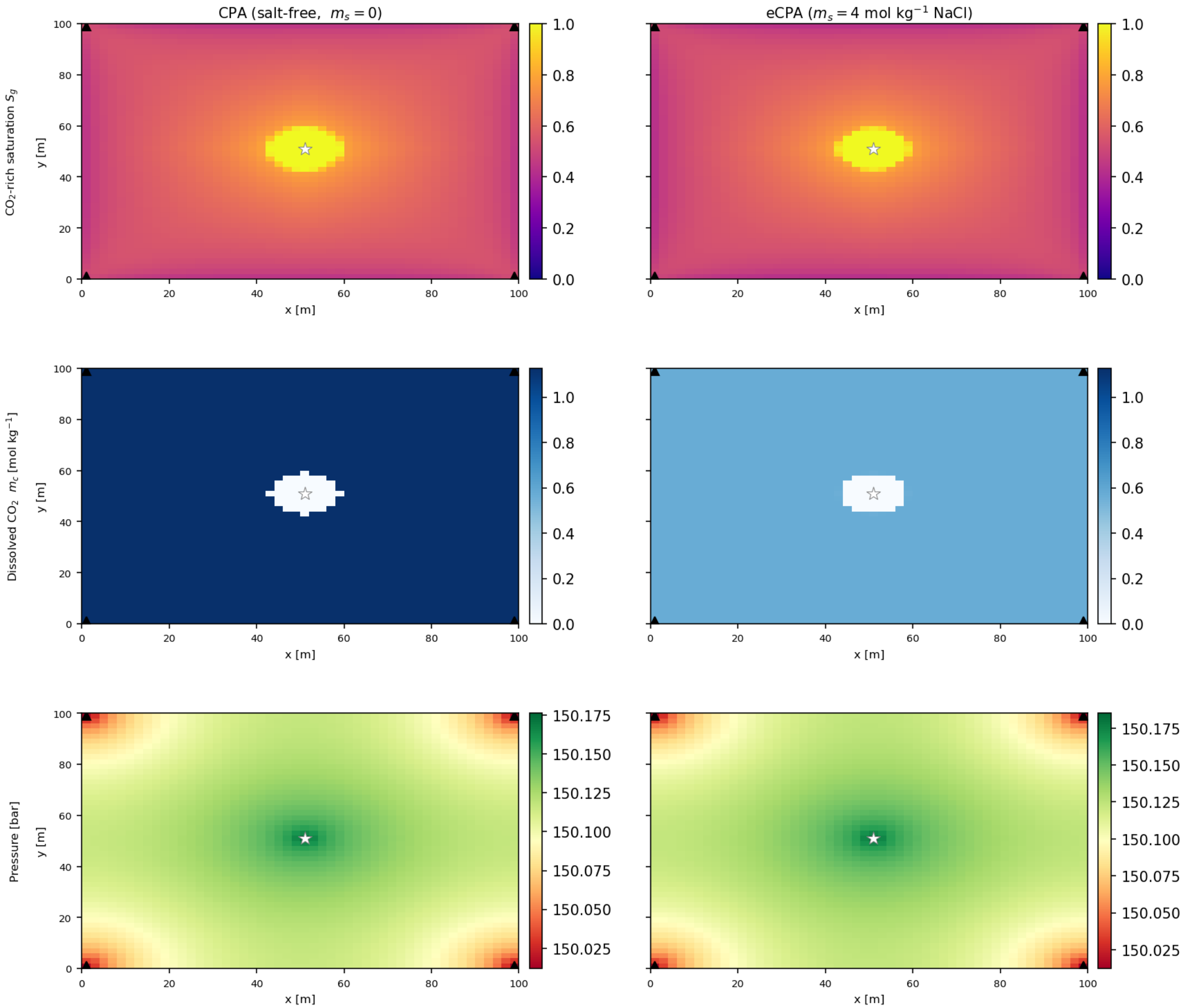


Figure S12: Final-state ($t = 5\,\mathrm{yr}$) spatial fields for the salt-free CPA case ($m_s = 0$, left column) and the saline eCPA case ($m_s = 4\,\mathrm{mol\,kg^{-1}}$ NaCl, right column). *Top row*: CO_2-rich phase saturation S_g. The plumes are nearly identical, reflecting advection-dominated displacement at 20% PVI yr^{-1}. *Middle row*: dissolved CO_2 molality m_c in the aqueous phase. Inside the plume the aqueous phase is CO_2-saturated at the eCPA equilibrium value: $\approx$1.25 mol kg^{-1} (salt-free) vs. $\approx$0.63 mol kg^{-1} ($m_s = 4\,\mathrm{mol\,kg^{-1}}$), a factor-of-two reduction from the salting-out effect. The white region near the central injector (white star) marks cells where brine has been completely expelled into the CO_2-rich phase (drying-out). *Bottom row*: pressure field; the $\approx$0.15 bar gradient across the 100 m domain reflects the small pressure drop required to drive 20% PVI yr^{-1} through 100 mD rock with BHP-controlled producers at 150 bar. Black triangles: producers.

While the saturation fields are similar, the implications for long-term CO_2 storage are markedly different. Solubility trapping—the dissolution of CO_2 into the formation brine—is a key mechanism for storage permanence, because dissolved CO_2 cannot escape as a buoyant supercritical phase. Fig. S13 shows the fraction of the in-situ CO_2 inventory that is dissolved in brine (solubility trapping, solid lines) versus residing in the CO_2-rich phase (structural trapping, dashed lines) as a function of time for both cases. The gap between the CPA and eCPA curves is clear and persistent throughout: at $t = 5\,\mathrm{yr}$ the solubility trapping fraction is roughly a factor of two lower for the saline case, directly reflecting the reduction in equilibrium CO_2 solubility from the salting-out effect. This demonstrates that brine salinity has a quantitatively significant impact on long-term storage permanence even when the short-term displacement dynamics appear unaffected.

Flash performance in the simulator

At every time step, the stability test and flash are called at each of the $N = 2500$ grid cells. Cells identified as trivially single-phase bypass the flash entirely; the remaining two-phase candidates are dispatched in parallel across all available CPU cores. As the plume grows, an increasing fraction of the domain becomes two-phase until essentially all cells require a flash call (peak ≈2500 calls per step). Performance numbers for both simulations are summarized in Table F.13; the flash accounts for the dominant fraction of total step cost throughout both runs, confirming that the algorithmic efficiency of the warm-started flash directly governs simulation throughput. Interestingly, the eCPA flash achieves a higher throughput than CPA (14,270 vs. 8,840 $\mathrm{calls\,s^{-1}}$), even though the eCPA EoS involves more unknowns and is inherently more expensive per iteration. While one would expect CPA to be the faster of the two in general, the solution-table warm start appears to provide particularly accurate initial guesses for the eCPA system at the temperature, pressure, and salinity conditions of these simulations ($T = 77\,^\circ\mathrm{C}$, $P \approx 150\,\mathrm{bar}$, $m_s = 4\,\mathrm{mol\,kg^{-1}}$), resulting in fewer SSI iterations to convergence than for the salt-free CPA case at otherwise identical conditions.

Table F.13: Aggregate flash performance for the 5-year simulations (50×50 grid, $T = 77\,^\circ\mathrm{C}$, $P \approx 150\,\mathrm{bar}$, 300 time steps, 48 CPU cores). Total wall time: 91 s (CPA) and 63 s (eCPA).

Metric	CPA ($m_s = 0$)	eCPA ($m_s = 4$)
Grid cells	2500	2500
Time steps	300	300
Mean two-phase cells / step	2219 (max 2479)	2213 (max 2479)
Mean flash wall time / step (s)	0.247	0.151
Flash fraction of step cost	87%	80%
Flash throughput ($\mathrm{calls\,s^{-1}}$)	8,840	14,270

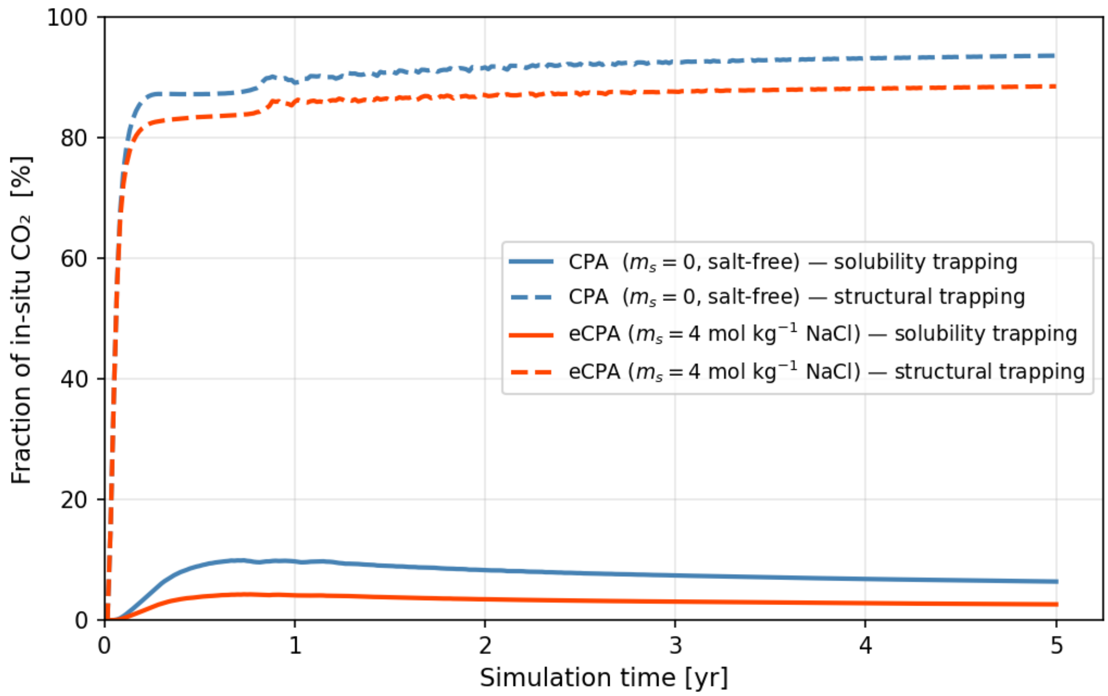


Figure S13: Fraction of total in-situ CO_2 inventory undergoing solubility trapping (dissolved in the aqueous phase, solid lines) versus structural trapping (residing in the CO_2-rich phase, dashed lines) as a function of time. Blue: salt-free brine (CPA, $m_s = 0$); red: saline brine (eCPA, $m_s = 4\,\mathrm{mol\,kg^{-1}}$ NaCl). The persistent factor-of-two gap in solubility trapping between the two cases reflects the reduction in equilibrium CO_2 solubility due to the salting-out effect, which has negligible impact on short-term displacement dynamics but is critical for long-term storage permanence.